

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\160-C-Vertical.pvdb

Date of Analysis : Sep 9, 2026 9:37am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User

FileName : 160-C-Vertical

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Vessel Design Summary:

Step: 36

9:37am

Sep 9, 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 247.650 mm ID	
Vessel Design Length, Tangent to Tangent	3504.35 mm
Distance of Bottom Tangent above Grade	300.00 mm
Specified Datum Line Distance	0.00 mm
Shell Side Design Temperature	430 °C
Channel Side Design Temperature	270 °C
Shell Side Design Pressure	5900.000 KPa.
Channel Side Design Pressure	2400.000 KPa.
Shop Shell Side Test Pressure	13058.051 KPa.
Field Shell Side Test Pressure	7670.000 KPa.
Shop Channel Side Test Pressure	6449.185 KPa.
Field Channel Side Test Pressure	3120.000 KPa.
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-335 P11	...	...	K11597	
Flange	SA-336 F11	2	...	K11572	
Flange	SA-335 P11	...	...	K11597	
Nozzle	SA-335 P11	...	...	K11597	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Sup. Lugs	SA-387 11	2	...	K11789	
Tubes	SA-213 T11	...	...	K11597	
Tubesheet	SA-387 11	2	...	K11789	
Flg Bolting	SA-193 B16	...	<= 64	K14072	
Lug Bolting	SA-193 B16	...	<= 64	K14072	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Blind flange	2434.473	103.42	5023.000	3.0000	N/A	No
Bottom ch flg	2433.983	103.42	6791.000	3.0000	N/A	No
Bottom ch shell	2433.140	103.42	6792.000	3.0000	N/A	No
Main Shell	5924.995	103.42	6072.000	3.0000	N/A	No
Top ch shell	2404.225	103.42	6821.000	3.0000	N/A	No
Top ch flg	2401.284	103.42	6248.000	3.0000	N/A	No
Top Blind Flang	2400.441	103.42	4060.000	3.0000	N/A	No

Liquid Level: 3417.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:



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Vessel Design Summary:

Step: 36 9:37am Sep 9,2026

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	48.2	45.0	53.0	45.0	37.0	...	1.00	1.00
Body Flg	127.2	79.0	62.0	55.0	46.7	...	1.00	1.00
Cylinder	427.2	300.0	12.7	11.1	5.8	...	1.00	1.00
Cylinder	3027.2	2550.0	12.7	11.1	10.9	...	1.00	1.00
Cylinder	3377.2	300.0	12.7	11.1	5.8	...	1.00	1.00
Body Flg	3459.4	79.0	62.0	55.0	51.2	...	1.00	1.00
Body Flg	3504.3	45.0	53.0	45.0	39.8	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	45	37.0586	103.419	No Calc
20	30	55	46.736	103.419	No Calc
30	Tsht	11.1125	3.93467	103.419	3791
Tsht	40	11.1125	No Calc	103.419	No Calc
40	Tsht	11.1125	5.44306	103.419	1969
Tsht	50	11.1125	No Calc	103.419	No Calc
50	60	11.1125	3.93474	103.419	3791
60	70	55	51.181	103.419	No Calc
70	80	45	39.8272	103.419	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	299.969	9394.02	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	2550	210101	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	300.031	9395.75	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Support Lugs	WRC-107/537	0.963	n/a	Passed
Support Lugs	WRC-107/537	0.560	n/a	Passed
N1-Tube Inlet	WRC-297	0.991	Shell	Passed
N3-Shell inlet	WRC-297	0.960	Shell	Passed
N4-Shell outle	WRC-297	0.960	Shell	Passed
N2-Tube Outlet	WRC-297	0.969	Shell	Passed

Loads for Foundation/Support Design:

Total Wind Shear on Support	97. Kg
Total Earthquake Shear on Support	75. Kg
Wind Moment on Support	0. Kg-m.
Earthquake Moment on Support	39. Kg-m.
Wind Load Case Test Moment on Support	0. Kg-m.



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Vessel Design Summary:

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Test Case Wind Shear on Support

48. Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	993.2 kg.
Shop Test - Fabricated + water (full)	1137.3 kg.
Shipping - Fab. + rem. intls.+ shipping app.	993.2 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	993.2 kg.
Empty - Fab. + intls. + details + wghts.	1229.1 kg.
Operating - Empty + operating liquid (no ca)	1372.9 kg.
Field Test - Empty weight + test medium (full)	1373.2 kg.

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Nozzle Summary:

Step: 34 9:37am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	6792	...	...	OK	5.82	OK	No Calc[*]
N3-Shell inlet	6072	OK	...	OK	8.26	OK	Passed
N4-Shell outlet	6072	...	...	OK	7.80	OK	No Calc[*]
N2-Tube Outlet		...	...	OK	5.79	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6072 Nozzle : N3-Shell inlet [Shellside]
 Minimum MAWP Nozzles : 6792 Nozzle : N1-Tube Inlet [Tubeside]

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1-Tube Inlet	277.175	270.000	88.352
40	N3-Shell inlet	977.175	270.000	196.099
40	N4-Shell outlet	2527.175	90.000	113.600
50	N2-Tube Outlet	3227.175	90.000	186.650

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 33 9:37am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Tube Inlet	2.000 in	XXS	LWN	2.375	11.074	...	...	164.84	600
N4-Shell outlet	2.000 in Actual		LWN	3.320	16.764	...	...	215.36	900
N3-Shell inlet	4.000 in	120	LWN	4.500	11.125	200.00	12.00	221.94	900
N2-Tube Outlet	4.000 in	160	WNF	4.500	13.487	214.30	16.00	175.09	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Tube Inl	SA-335 P11	11.074	10.000	...	...	...
N4-Shell ou	SA-335 P11	6.823	10.000	...	...	...
N3-Shell in	SA-335 P11	...	10.000	8.000	12.000	...
N2-Tube Out	SA-335 P11	8.112	10.000	12.000	16.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Tube Inlet	277.175	270.0	150.00	0.00	Bottom ch shell
N4-Shell outlet	2527.175	90.0	196.85	0.00	Main Shell
N3-Shell inlet	977.175	270.0	196.85	0.00	Main Shell
N2-Tube Outlet	3227.175	90.0	150.00	0.00	Top ch shell

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Bill of Material:

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 53.0mm THK X 430.0mm OD	SA-336 F11
1	WELD NECK FLANGE: 62.0mm THK X 430.0mm OD	SA-336 F11
2	CYLINDER: 12.7mm THK X 247.6mm ID X 300.0mm	SA-335 P11
1	CYLINDER: 12.7mm THK X 247.6mm ID X 2550.0mm	SA-335 P11
1	WELD NECK FLANGE: 62.0mm THK X 430.0mm OD	SA-335 P11
1	BLIND FLANGE: 53.0mm THK X 430.0mm OD	SA-335 P11
2	INSULATION: 50mm X 100mm THK	Hot insulation
2	INSULATION: 86mm X 100mm THK	Hot insulation
2	INSULATION: 300mm X 100mm THK	Hot insulation
7	BAFFLES: 286.0mm DIA X 6.0mm THK	SA-516 70
4	TIE RODS: 9mm DIA X 2110mm	SA-193 B7M
2	TUBESHEET: 269.3mm X 50.0mm THK	SA-387 11
60	TUBES: 2650.0mm X 19.0mm DIA X 2.1mm THK	SA-213 T11
1	INSULATION: 2550mm X 150mm THK	Hot insulation
4	GUSSET PLATES: 225.0mm X 75.0mm X 300.0mm X 25.0mm	SA-387 11
2	BOTTOM PLATES: 300.0mm X 225.0mm X 25.0mm	SA-387 11
2	TOP PLATES: 300.0mm X 75.0mm X 25.0mm	SA-387 11
2	SUPPORT LUG RE-PADS: 160.0mm X 350.0mm X 25.0mm	SA-387 11
2	LUG BOLTS: 30.0mm User Root Area	SA-193 B16
2	GASKET: 305.6mm OD X 279.6mm ID	...
24	BODY FLANGE BOLTS: 25.4mm DIA	SA-193 B16
48	NUTS FOR BODY FLANGE BOLTS: 25.4mm DIA	...
1	NAMEPLATE	...



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MDMT Summary:

Step: 35 9:37am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Main Shell [8]		B	-26	-48		0.398	11.113	1.00	Yes
N3-Shell inlet[1]		B	-29	-48		0.377	9.735	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
N4-Shell outle[1]		B	-26	-36		0.807	11.113	1.00	No
Nozzle Flg [5]		A	-18	-48					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No
Tubesheet: SS[13]		A	-1	-104		0.307	12.700	1.00	No

Warmest MDMT: -1 -36

Blind flange [11]	B	-25	-45		0.648	11.250	1.00	No
Bottom ch flg[11]	B	-26	-48		0.436	11.113	1.00	No
Top ch flg [11]	B	-26	-48		0.436	11.113	1.00	No
Top Blind Fla[11]	B	-25	-39		0.758	11.250	1.00	No
Bottom ch shel[8]	B	-26	-104		0.248	11.113	1.00	Yes
Top ch shell [8]	B	-26	-104		0.248	11.113	1.00	Yes
N1-Tube Inlet [1]	B	-29	-104		0.069	9.690	1.00	No
Nozzle Flg [5]	A	-18	-104					
N2-Tube Outlet[1]	B	-3	-104		0.109	11.801	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Tubesheet: CS[14]	A	-1	-104		0.139	12.700	1.00	No
Bolting [21]		-29						
Bolting [21]		-29						

Warmest MDMT: -1 -39

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-36.0	-12.0	Pass
Channel/Tube	-29.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.



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MDMT Summary:

Step: 35 9:37am Sep 9, 2026

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:37am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Note: Lug supports should be modeled on an element of their own whose length is just slightly greater than the height of the lug assembly. The vessel does not have to be constructed with this extra element, just modeled this way. This is to insure there is proper mass distribution for these types of models.

Warning:

This is a vertical vessel on support lugs and the earthquake type selected is a building code. Building codes assume all of the mass is located above the point of support. This makes their use on lug supported vessels questionable. It is strongly suggested that based on the seismic zone and applicable parameters, an appropriate lateral (Gx) and (+-) vertical (Gy) acceleration be applied using the G loading option.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:37am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	5900	KPa.
Channel Side Design Pressure	2400	KPa.
Shell Side Design Temperature	430.0	°C
Channel Side Design Temperature	270.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

Step: 1 9:37am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	92.7 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

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Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Blind flange
Distance "FROM" to "TO"	45 mm
Flange Outside Diameter	430 mm
Element Thickness	45 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	53 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2400 KPa.
Design Temperature Internal Pressure	270 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	247.5 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	16.002 mm
Yield Stress, Operating	222.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K11572
Class / Thickness / Grade	2 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	50 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 70



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Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	50	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Bottom ch flg
Distance "FROM" to "TO"	79 mm
Flange Inside Diameter	247.65 mm
Element Thickness	55 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	62 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2400 KPa.
Design Temperature Internal Pressure	270 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	86 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	86 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Bottom ch shell
Distance "FROM" to "TO"	300 mm
Inside Diameter	247.65 mm
Element Thickness	11.113 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	12.7 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2400 KPa.
Design Temperature Internal Pressure	270 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-335 P11
Allowable Stress, Ambient	118 N./mm ²
Allowable Stress, Operating	110.8 N./mm ²
Allowable Stress, Hydrotest	184.5 N./mm ²
Material Density	0.00775 kg./cm ³



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Input Echo:

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P Number Thickness	16.002	mm
Yield Stress, Operating	166.6	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K11597	
Product Form	Smls. pipe	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	300	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	300	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1-Tube Inlet	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	7.1458	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-335 P11	

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	2550	mm
Inside Diameter	247.65	mm
Element Thickness	11.113	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	12.7	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	5900	KPa.
Design Temperature Internal Pressure	430	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	430	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-335 P11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm



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Height/Length of Liquid	2550	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2550	mm
Thickness of Insulation	150	mm
Density	0.000352	kg./cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N3-Shell inlet	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	900	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	30.94	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-335 P11	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N4-Shell outlet	
Dist. from "FROM" Node / Offset dist	2050	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.656	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-335 P11	

Element From Node	40	
Detail Type	Lug	
Detail ID	Support Lugs	
Dist. from "FROM" Node / Offset dist	1276	mm
Number of Lugs	2	
Dist. from OD to Lug Cntrline(dlug)	250	mm
Height of Gusset Plates (hgp)	300	mm
Force Bearing Width (wfb)	80	mm
Weight of Lug	44.077	Kgf
Lug Start Angle (degrees)	45.0	

Element#	5/7	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Top ch shell	
Distance "FROM" to "TO"	300	mm
Inside Diameter	247.65	mm
Element Thickness	11.113	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	12.7	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2400	KPa.
Design Temperature Internal Pressure	270	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	270	°C



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Effective Diameter Multiplier	1.2
Material Name	SA-335 P11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	300 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	300 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³
Element From Node	50
Detail Type	Nozzle
Detail ID	N2-Tube Outlet
Dist. from "FROM" Node / Offset dist	150 mm
Nozzle Diameter	4 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	27.22 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-335 P11

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Flange
Description	Top ch flg
Distance "FROM" to "TO"	79 mm
Flange Inside Diameter	247.65 mm
Element Thickness	55 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	62 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2400 KPa.
Design Temperature Internal Pressure	270 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-335 P11
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	86 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	60
Detail Type	Insulation



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Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	86 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Flange
Description	Top Blind Flang
Distance "FROM" to "TO"	45 mm
Flange Outside Diameter	430 mm
Element Thickness	45 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	53 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2400 KPa.
Design Temperature Internal Pressure	270 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	270 °C
Effective Diameter Multiplier	1.2
Material Name	SA-335 P11
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	45 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	70
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	50 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³



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XY Coordinate Calculations:

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Blind flange		...	48.175	...	45
Bottom ch flg		...	127.175	...	79
Bottom ch shell		...	427.175	...	300
Main Shell		...	3027.18	...	2550
Top ch shell		...	3377.17	...	300
Top ch flg		...	3459.35	...	79
Top Blind Flang		...	3504.35	...	45

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: New Flange

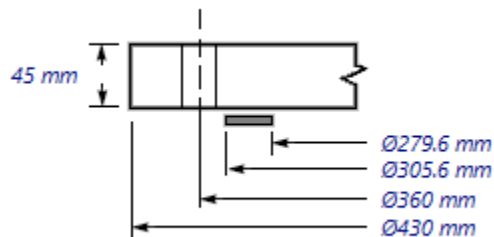
Flng: 9 9:37am Sep 9,2026

Flange Input Data Values

Description: New Flange :

Blind flange

Flange Type		Blind
Design Pressure	P	2434.47 KPa.
Design Temperature		270 °C
Internal Corrosion Allowance	ci	3.0000 mm
External Corrosion Allowance	ce	0.0000 mm
Use Corrosion Allowance in Thickness Calcs.		Yes
Flange Outside Diameter	A	430.000 mm
Flange Thickness	t	45.0000 mm
Flange Material		SA-336 F11
Flange Material UNS number		K11572
Flange Allowable Stress At Temperature	Sfo	138.00 N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00 N./mm ²
Bolt Material		SA-193 B16
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Perimeter along the Center of the Bolts	L	1130.973 mm
Diameter of Bolt Circle	C	360.000 mm
Nominal Bolt Diameter	a	25.4000 mm
Type of Threads		TEMA Thread Series
Number of Bolts		12
Flange Face Outside Diameter	Fod	316.650 mm
Flange Face Inside Diameter	Fid	247.650 mm
Flange Facing Sketch		Sketch 1a
Gasket Outside Diameter	Go	305.650 mm
Gasket Inside Diameter	Gi	279.650 mm
Gasket Factor	m	3.0000
Gasket Design Seating Stress	y	68.95 N./mm ²
Gasket Thickness	tg	3.1750 mm
Raised face depth	Fd	0.0000 mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	292.801 mm

Basic Flange and Bolt Loads:



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Flg Calc [Int P]: New Flange

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Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 293^2 \cdot 2434$$

$$= 16716.145 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 293 \cdot 3 \cdot 2434$$

$$= 8802.752 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(16716 + 8803 + 0, 0)$$

$$= 25518.898 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 293 + 0 \cdot 0 \cdot 0$$

$$= 41550.023 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((25519 + 0 + 4 \cdot 0 / 293) / 172, 41550 / 172)$$

$$= 23.691 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	23.691	42.658	
Radial Distance between Bolts and Edge:	26.987	35.000	
Circ. Spacing between the Bolts:	57.150	93.175	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 42.7 \cdot 172 / (68.9 \cdot \pi (306 + 280))$$

$$= 5.787 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 42.7$$

$$= 74815.61 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 25518.90 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (360 - 293) / 2$$

$$= 33.5996 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 25519 \cdot 33.6 \cdot 1 / (42^2 \cdot 293) + 0.3 \cdot 1 \cdot 2434 \cdot 293^2 / 42^2$$

$$= 66.43 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 74816 \cdot 33.6 \cdot 1 / (42^2 \cdot 293)$$



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Flg Calc [Int P]: New Flange

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= 90.69 N./mm²

Bolt Stress, Operating [BS_o]:

= W_{m1} / A_b

= 25519/42.7

= 58.67 N./mm²

Bolt Stress, Seating [BS_a]:

= (W_{m2} / A_b)

= (41550/42.7)

= 95.52 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	66.43	138.00	90.69	138.00
Bolting	58.67	172.00	95.52	172.00

Reqd. Blind Flange Thickness at Center 37.047 mm

Reqd. Blind Flange Thickness at Gasket 37.047 mm

Estimated M.A.W.P. (Operating) 5057 KPa.

Estimated Finished Weight of Flange at given Thk. 50.6 kg.

Estimated Unfinished Weight of Forging at given Thk 50.6 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.65, Temperature Reduction per Fig. UCS 66.1 = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -25 °C

Min Metal Temp. at Required thickness (UCS 66.1) -45 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	57.87	138.00	79.00	138.00
Bolting	58.67	172.00	95.52	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 34.061 mm

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Flg Calc [Int P]: Bottom ch fl

Flng: 10

9:37am

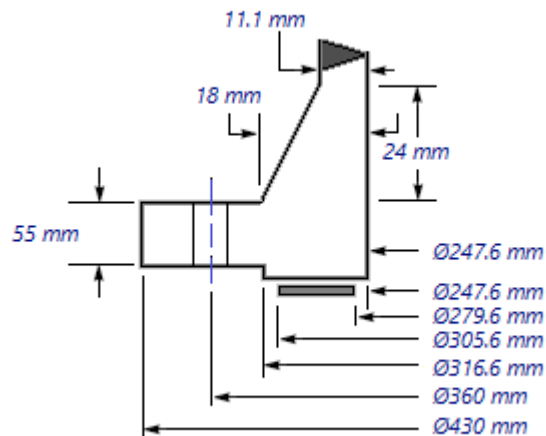
Sep 9,2026

Flange Input Data Values

Description: Bottom ch fl :

Bottom ch flg

Flange Type	Integral Weld Neck		
Design Pressure	P	2433.98	KPa.
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	247.650	mm
Flange Outside Diameter	A	430.000	mm
Flange Thickness	t	55.0000	mm
Thickness of Hub at Small End	go	11.1125	mm
Thickness of Hub at Large End	gl	18.0000	mm
Length of Hub	h	24.0000	mm
Flange Material		SA-336 F11	
Flange Material UNS number		K11572	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B16	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	360.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		12	
Flange Face Outside Diameter	Fod	316.650	mm
Flange Face Inside Diameter	Fid	247.650	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	305.650	mm
Gasket Inside Diameter	Gi	279.650	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: Bottom ch fl Flng: 10 9:37am Sep 9,2026

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2434 \cdot (248/2 + 3)) / (111 \cdot 1 \cdot 0.6 \cdot 2434) + Ca$$

$$= 5.8234 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (111 \cdot 1 \cdot 8.11) / (127 + 0.6 \cdot 8.11)$$

$$= 6825.049 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 11.1) / (124 + 0.6 \cdot 11.1)$$

$$= 10047.897 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	52.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	253.650 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	15.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	8.113 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	38.175 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $b_o = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	292.801 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 293^2 \cdot 2434$$

$$= 16712.779 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 293 \cdot 3 \cdot 2434$$

$$= 8800.979 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 254^2 \cdot 2434$$

$$= 12542.197 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 16713 - 12542$$

$$= 4170.583 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(16713 + 8801 + 0, 0)$$

$$= 25513.760 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 293 + 0 \cdot 0 \cdot 0$$

$$= 41550.023 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + Fa + 4Me/G)/S_b, W_g/S_a)$$

$$= \max((25514 + 0 + 4 \cdot 0/293)/172, 41550/172)$$

$$= 23.691 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 52/(3 + 0.5)$$

$$= 139.943 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 360 \cdot \sin(\pi/12)$$

$$= 93.175 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{93.2/(2 \cdot 25.4 + 52)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	23.691	42.658	
Radial Distance between Hub and Bolts:	34.925	38.175	
Radial Distance between Bolts and Edge:	26.987	35.000	
Circ. Spacing between the Bolts:	57.150	93.175	139.943

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (\gamma \cdot \pi (Go + Gi))$$

$$= 42.7 \cdot 172 / (68.9 \cdot \pi (306 + 280))$$

$$= 5.787 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 42.7$$

$$= 74815.61 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 25514 - 16713$$

$$= 8800.98 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (360 - 293) / 2$$

$$= 33.5996 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((360 - 254) / 2 + 33.6)$$

$$= 43.3873 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (360 - 254 - 15) / 2$$

$$= 45.6750 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	12542.	45.6750	1.0000	573.
Face Pressure, Mt	4171.	43.3873	1.0000	181.
Gasket Load, Mg	8801.	33.5996	1.0000	296.
Gasket Seating, Matm	74816.	33.5996	1.0000	2514.



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Total Moment for Operation, Mo 1050. Kg-m.
Total Moment for Gasket seating, Mg 2514. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 45.362 mm
Hub Ratio, h/h0 = HL / H0 0.529
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.849

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.823
Factor V per 4.16.5 0.241
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 430/254
= 1.695

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K)) / (K² - 1))/(K - 1))
= (0.66845 + (5.7169(1.7² • log10(1.7))/(1.7² - 1))/(1.7 - 1))
= 3.852

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
= (1.7²(1 + 8.55246 • log10(1.7)) - 1)/((1.0472 + 1.9448 • 1.7²) • (1.7 - 1))
= 1.627

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
= (1.7²(1 + 8.55246 • log10(1.7)) - 1)/(1.36136(1.7² - 1) • (1.7 - 1))
= 4.233

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
= (1.7² + 1)/(1.7² - 1)
= 2.067

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 4.23 • 8.11² • 45.4/0.24
= 52529.477 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.82/45.4
= 0.018 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (52 • 0.018 + 1)/1.63 + 52³/52529
= 3.871

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1•1050/254)/(3.87•15²)
= 46.59 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
= (1•2514/254)/(3.87•15²)
= 111.59 N./mm²



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Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.26 \cdot 1050 / 254) / (3.87 \cdot 52^2)$$

$$= 8.75 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.26 \cdot 2514 / 254) / (3.87 \cdot 52^2)$$

$$= 20.96 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (3.85 \cdot 1050 / (52^2 \cdot 254)) - 2.07 \cdot 8.75$$

$$= 39.72 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (3.85 \cdot 2514 / (52^2 \cdot 254)) - 2.07 \cdot 21$$

$$= 95.13 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (46.6 + \max(8.75, 39.7)) / 2$$

$$= 43.15 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (112 + \max(21, 95.1)) / 2$$

$$= 103.36 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 25514 / 42.7$$

$$= 58.65 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (41550 / 42.7)$$

$$= 95.52 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	46.59	207.00	111.59	207.00
Radial Flange	8.75	138.00	20.96	138.00
Tangential Flange	39.72	138.00	95.13	138.00
Maximum Average	43.15	138.00	103.36	138.00
Bolting	58.65	172.00	95.52	172.00

Minimum Required Flange Thickness 46.736 mm

Estimated M.A.W.P. (Operating) 6825 KPa.

Estimated Finished Weight of Flange at given Thk. 43.6 kg.

Estimated Unfinished Weight of Forging at given Thk 59.4 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

	Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub Sh	411.99	508.80	0.810	411.99	549.99	0.749



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Tangential Flange	St	351.20	381.60	0.920	351.20	412.49	0.851
Tangential Flange	Sto	186.77	254.40	0.734	186.77	275.00	0.679
Combined	Sh/f + St	763.20	763.20	1.000	763.20	824.99	0.925
Combined	Sr + St	428.60	763.20	0.562	428.60	824.99	0.520
Combined	Sh + Sto	598.76	763.20	0.785	598.76	824.99	0.726
Radial Flange	Sr	77.40	508.80	0.152	77.40	549.99	0.141
Maximum Stress Ratio				1.000			0.925

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.63/1.7)

= 1.0000

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)

= max(0.25; 0.85/1.78) if h/ho < 0.35, else max (0.25; 0.35/0.76)

= 0.4533

In this case, the hub ratio h/ho, 0.5291 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:



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Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	635.0 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	3.5548 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	275.0 N./mm ²
Flange Material Operating Yield Stress	Syo	222.6 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.223 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (306^2 - 280^2)$$

$$= 119.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 120 / (12 \cdot 3.55)$$

$$= 580.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(580; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 635)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 508 \geq 140 (120 / (12 \cdot 3.55))$$

$$:: 507.5 \geq 392.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 508 \geq (97 \cdot 120 + \pi/4 \cdot 2434 \cdot 280^2) / (0.7 \cdot 3.55 \cdot 12)$$

$$:: 507.5 \geq 438.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 508 \leq 276 (120 / (12 \cdot 3.55))$$

$$:: 507.5 \leq 773.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetaamax)$$

$$:: 508 \leq 635 (1/0.22)$$

$$:: 507.5 \leq 2846.4 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$



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$$= 508 \cdot 0.2 \cdot 3.55 \cdot 25.4$$

$$= 93.5 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 N./mm²
 Bolt Allowable Stress for: SA-193 B16 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2514 / 1 \cdot 9807 \cdot 0.24 / (2.69 \cdot 204092 \cdot 8.11^2 \cdot 45.4 \cdot 0.3)$$

$$= 0.628 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1050 / 1 \cdot 9807 \cdot 0.24 / (2.69 \cdot 188304 \cdot 8.11^2 \cdot 45.4 \cdot 0.3)$$

$$= 0.284 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2514 / 1 \cdot 9807 \cdot 0.24 / (3.87 \cdot 204092 \cdot 8.11^2 \cdot 45.4 \cdot 0.3)$$

$$= 0.437 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1050 / 1 \cdot 9807 \cdot 0.24 / (3.87 \cdot 188304 \cdot 8.11^2 \cdot 45.4 \cdot 0.3)$$

$$= 0.197 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.44, Temperature Reduction per Fig. UCS 66.1 = 43 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	42.98	207.00	100.68	207.00
Radial Flange	10.05	138.00	23.55	138.00
Tangential Flange	31.84	138.00	74.57	138.00
Maximum Average	37.41	138.00	87.62	138.00
Bolting	58.65	172.00	95.52	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 40.132 mm

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Flg Calc [Int P]: New Flange

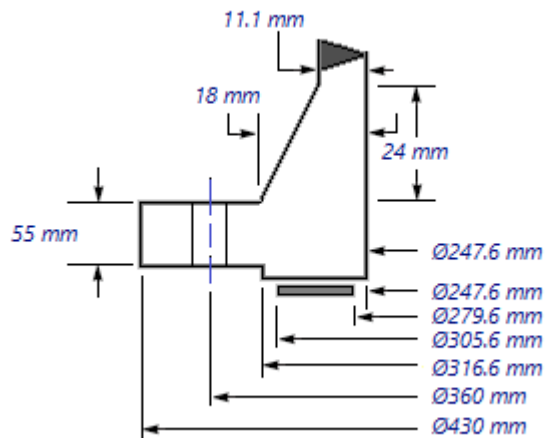
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Flange Input Data Values

Description: New Flange :

Top ch flg

Flange Type	Integral Weld Neck		
Design Pressure	P	2401.28	KPa.
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	247.650	mm
Flange Outside Diameter	A	430.000	mm
Flange Thickness	t	55.0000	mm
Thickness of Hub at Small End	go	11.1125	mm
Thickness of Hub at Large End	gl	18.0000	mm
Length of Hub	h	24.0000	mm
Flange Material	SA-335 P11		
Flange Material UNS number	K11597		
Flange Allowable Stress At Temperature	Sfo	110.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	118.00	N./mm ²
Bolt Material	SA-193 B16		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	360.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	12		
Flange Face Outside Diameter	Fod	316.650	mm
Flange Face Inside Diameter	Fid	247.650	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	305.650	mm
Gasket Inside Diameter	Gi	279.650	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2401 \cdot (248/2 + 3)) / (111 \cdot 1 \cdot 0.6 \cdot 2401) + Ca$$

$$= 5.7850 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (111 \cdot 1 \cdot 8.11) / (127 + 0.6 \cdot 8.11)$$

$$= 6825.049 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 11.1) / (124 + 0.6 \cdot 11.1)$$

$$= 10047.897 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	52.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	253.650 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	15.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	8.113 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	38.175 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	292.801 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 293^2 \cdot 2401$$

$$= 16488.258 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 293 \cdot 3 \cdot 2401$$

$$= 8682.744 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 254^2 \cdot 2401$$

$$= 12373.703 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 16488 - 12374$$

$$= 4114.554 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(16488 + 8683 + 0, 0)$$

$$= 25171.000 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 293 + 0 \cdot 0 \cdot 0$$

$$= 41550.023 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((25171 + 0 + 4 \cdot 0/293)/172, 41550/172)$$

$$= 23.691 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 52/(3 + 0.5)$$

$$= 139.943 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 360 \cdot \sin(\pi / 12)$$

$$= 93.175 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{93.2/(2 \cdot 25.4 + 52)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	23.691	42.658	
Radial Distance between Hub and Bolts:	34.925	38.175	
Radial Distance between Bolts and Edge:	26.987	35.000	
Circ. Spacing between the Bolts:	57.150	93.175	139.943

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 42.7 \cdot 172 / (68.9 \cdot \pi (306 + 280))$$

$$= 5.787 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 42.7$$

$$= 74815.61 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 25171 - 16488$$

$$= 8682.74 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (360 - 293) / 2$$

$$= 33.5996 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((360 - 254) / 2 + 33.6)$$

$$= 43.3873 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (360 - 254 - 15) / 2$$

$$= 45.6750 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	12374.	45.6750	1.0000	565.
Face Pressure, Mt	4115.	43.3873	1.0000	179.
Gasket Load, Mg	8683.	33.5996	1.0000	292.
Gasket Seating, Matm	74816.	33.5996	1.0000	2514.



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Flg Calc [Int P]: New Flange

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Total Moment for Operation, Mo 1035. Kg-m.
Total Moment for Gasket seating, Mg 2514. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 45.362 mm
Hub Ratio, h/h0 = HL / H0 0.529
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.849

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.823
Factor V per 4.16.5 0.241
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 430/254
= 1.695

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.7² • log10(1.7))) / (1.7² - 1))/(1.7 - 1)
= 3.852

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
= (1.7²(1 + 8.55246 • log10(1.7)) - 1)/((1.0472 + 1.9448 • 1.7²) • (1.7 - 1))
= 1.627

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
= (1.7²(1 + 8.55246 • log10(1.7)) - 1)/(1.36136(1.7² - 1) • (1.7 - 1))
= 4.233

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
= (1.7² + 1)/(1.7² - 1)
= 2.067

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 4.23 • 8.11² • 45.4/0.24
= 52529.477 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.82/45.4
= 0.018 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (52 • 0.018 + 1)/1.63 + 52³/52529
= 3.871

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1•1035/254)/(3.87•15²)
= 45.96 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
= (1•2514/254)/(3.87•15²)
= 111.59 N./mm²



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Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.26 \cdot 1035 / 254) / (3.87 \cdot 52^2)$$

$$= 8.63 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.26 \cdot 2514 / 254) / (3.87 \cdot 52^2)$$

$$= 20.96 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.85 \cdot 1035 / (52^2 \cdot 254)) - 2.07 \cdot 8.63$$

$$= 39.18 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3.85 \cdot 2514 / (52^2 \cdot 254)) - 2.07 \cdot 21$$

$$= 95.13 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (46 + \max(8.63, 39.2)) / 2$$

$$= 42.57 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (112 + \max(21, 95.1)) / 2$$

$$= 103.36 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 25171 / 42.7$$

$$= 57.87 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (41550 / 42.7)$$

$$= 95.52 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	45.96	166.20	111.59	177.00
Radial Flange	8.63	110.80	20.96	118.00
Tangential Flange	39.18	110.80	95.13	118.00
Maximum Average	42.57	110.80	103.36	118.00
Bolting	57.87	172.00	95.52	172.00

Minimum Required Flange Thickness

51.181 mm

Estimated M.A.W.P. (Operating)

6249 KPa.

Estimated Finished Weight of Flange at given Thk.

43.6 kg.

Estimated Unfinished Weight of Forging at given Thk

59.4 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2514 / 1 \cdot 9807 \cdot 0.24 / (3.28 \cdot 204092 \cdot 8.11^2 \cdot 45.4 \cdot 0.3)$$

$$= 0.516 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:



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$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1035 / 1 \cdot 9807 \cdot 0.24 / (3.28 \cdot 188304 \cdot 8.11^2 \cdot 45.4 \cdot 0.3)$$

$$= 0.230 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2514 / 1 \cdot 9807 \cdot 0.24 / (3.87 \cdot 204092 \cdot 8.11^2 \cdot 45.4 \cdot 0.3)$$

$$= 0.437 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 1035 / 1 \cdot 9807 \cdot 0.24 / (3.87 \cdot 188304 \cdot 8.11^2 \cdot 45.4 \cdot 0.3)$$

$$= 0.195 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.44, Temperature Reduction per Fig. UCS 66.1 = 43 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	42.41	177.00	100.68	177.00
Radial Flange	9.92	118.00	23.55	118.00
Tangential Flange	31.41	118.00	74.57	118.00
Maximum Average	36.91	118.00	87.62	118.00
Bolting	57.87	172.00	95.52	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 45.009 mm

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Flg Calc [Int P]: New Flange

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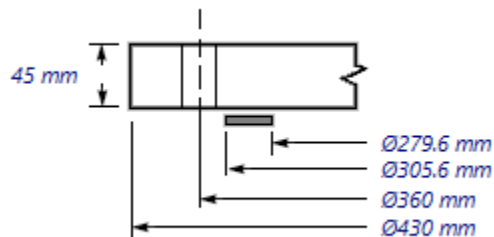
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Flange Input Data Values

Description: New Flange :

Top Blind Flang

Flange Type		Blind
Design Pressure	P	2400.44 KPa.
Design Temperature		270 °C
Internal Corrosion Allowance	ci	3.0000 mm
External Corrosion Allowance	ce	0.0000 mm
Use Corrosion Allowance in Thickness Calcs.		Yes
Flange Outside Diameter	A	430.000 mm
Flange Thickness	t	45.0000 mm
Flange Material		SA-335 P11
Flange Material UNS number		K11597
Flange Allowable Stress At Temperature	Sfo	110.80 N./mm ²
Flange Allowable Stress At Ambient	Sfa	118.00 N./mm ²
Bolt Material		SA-193 B16
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Perimeter along the Center of the Bolts	L	1130.973 mm
Diameter of Bolt Circle	C	360.000 mm
Nominal Bolt Diameter	a	25.4000 mm
Type of Threads		TEMA Thread Series
Number of Bolts		12
Flange Face Outside Diameter	Fod	316.650 mm
Flange Face Inside Diameter	Fid	247.650 mm
Flange Facing Sketch		Sketch 1a
Gasket Outside Diameter	Go	305.650 mm
Gasket Inside Diameter	Gi	279.650 mm
Gasket Factor	m	3.0000
Gasket Design Seating Stress	y	68.95 N./mm ²
Gasket Thickness	tg	3.1750 mm
Raised face depth	Fd	0.0000 mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb * \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = Go - 2 * b$	292.801 mm

Basic Flange and Bolt Loads:



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Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 293^2 \cdot 2400$$

$$= 16482.467 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 293 \cdot 3 \cdot 2400$$

$$= 8679.696 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(16482 + 8680 + 0, 0)$$

$$= 25162.162 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 293 + 0 \cdot 0 \cdot 0$$

$$= 41550.023 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((25162 + 0 + 4 \cdot 0 / 293) / 172, 41550 / 172)$$

$$= 23.691 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	23.691	42.658	
Radial Distance between Bolts and Edge:	26.987	35.000	
Circ. Spacing between the Bolts:	57.150	93.175	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 42.7 \cdot 172 / (68.9 \cdot \pi (306 + 280))$$

$$= 5.787 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 42.7$$

$$= 74815.61 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 25162.16 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (360 - 293) / 2$$

$$= 33.5996 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 25162 \cdot 33.6 \cdot 1 / (42^2 \cdot 293) + 0.3 \cdot 1 \cdot 2400 \cdot 293^2 / 42^2$$

$$= 65.50 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 74816 \cdot 33.6 \cdot 1 / (42^2 \cdot 293)$$



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Flg Calc [Int P]: New Flange

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= 90.69 N./mm²

Bolt Stress, Operating [BSO]:

= Wm1 / Ab

= 25162/42.7

= 57.85 N./mm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)

= (41550/42.7)

= 95.52 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	65.50	110.80	90.69	118.00
Bolting	57.85	172.00	95.52	172.00

Reqd. Blind Flange Thickness at Center 39.820 mm

Reqd. Blind Flange Thickness at Gasket 39.820 mm

Estimated M.A.W.P. (Operating) 4061 KPa.

Estimated Finished Weight of Flange at given Thk. 50.6 kg.

Estimated Unfinished Weight of Forging at given Thk 50.6 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.76, Temperature Reduction per Fig. UCS 66.1 = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -25 °C

Min Metal Temp. at Required thickness (UCS 66.1) -39 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	57.06	118.00	79.00	118.00
Bolting	57.85	172.00	95.52	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 36.830 mm

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Internal Pressure Calculations: Step: 7 9:37am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Blind flange		2434.5	53	3	430	138	1
Bottom ch flg		2434	62	3	247.65	138	2
Bottom ch shell		2433.1	12.7	3	247.65	110.8	3
Main Shell		5925	12.7	3	247.65	98.982	4
Top ch shell		2404.2	12.7	3	247.65	110.8	5
Top ch flg		2401.3	62	3	247.65	110.8	6
Top Blind Flang		2400.4	53	3	430	110.8	7

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Blind flange		2400	5023	5806	45	37.0473	1
Bottom ch flg		2400	6791	7137	55	46.736	2
Bottom ch shell		2400	6792	10048	11.1125	5.82241	3
Main Shell		5900	6072	10048	11.1125	10.875	4
Top ch shell		2400	6821	10048	11.1125	5.78843	5
Top ch flg		2400	6248	7137	55	51.181	6
Top Blind Flang		2400	4060	4964	45	39.8197	7

Warning: Channel Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 6072 KPa.
 Shell Side MAPnc = 10048 KPa.
 Channel Side MAWP = 3764 KPa.
 Channel Side MAPnc = 4964 KPa.

Elements Suitable for Design Internal Pressure.

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Element: Bottom ch shell

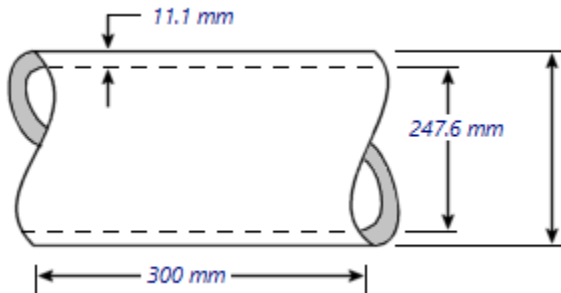
Step: 8

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom ch shell

Cylindrical Shell Design Information:			
Design Pressure P	2433.140 KPa.	Int. Design Temperature	270 °C
Material	SA-335 P11	Ext. Design Temperature	270 °C
Length	300.000 mm	External Pressure Chart	CS-2
		UNS Number	K11597
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	110.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	247.650 mm	Actual Stress	39.5 N./mm ²
Spec. Min. Thk t,e, or sa	11.1125 mm	Spec. Nominal Thk.	12.700 mm
Surface Area	2573.4 cm ²	SG of Contents	1.000
Element Volume	14453.3 cm ³	Weight of Contents	14.4 Kg
Empty Weight	24.2 Kg	Operating Weight	38.6 Kg
		Comp. All. Ope. Stress	116.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2433 \cdot 127) / (111 \cdot 1 - 0.6 \cdot 2433) \\
 &= 2.8224 + 3.0000 = 5.8224 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 33.140 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (111 \cdot 1 \cdot 8.11) / (127 + 0.6 \cdot 8.11) \\
 &= 6825 - 33.1 = 6792 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118 \cdot 1 \cdot 11.1) / (124 + 0.6 \cdot 11.1) \\
 &= 10048 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2433 \cdot (127 + 0.6 \cdot 8.11)) / (1 \cdot 8.11) \\
 &= 39.500 \text{ N./mm}^2
 \end{aligned}$$



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 Element: Bottom ch shell Step: 8 9:37am Sep 9,2026

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 11.1, tr = 2.01, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.25, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3764.250 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	11.113 mm	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.248	Temperature Difference	78 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Bottom ch shell

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 270 °C

Elastic Modulus from Chart: CS-2 at 270 °C: 184606480 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.113	269.88	299.97	33.27	1.1115	0.0064678	94.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*94.6)/(3*33.3) = 3791 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
0.935	269.88	299.97	288.74	1.1115	0.0002426	22.40

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.4)/(3*289) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.113	269.88	9394.02	33.27	34.8088	0.0010199	67.78

$$MAEP = (4*B)/(3*(Do/t)) = (4*67.8)/(3*33.3) = 2717 \text{ KPa.}$$



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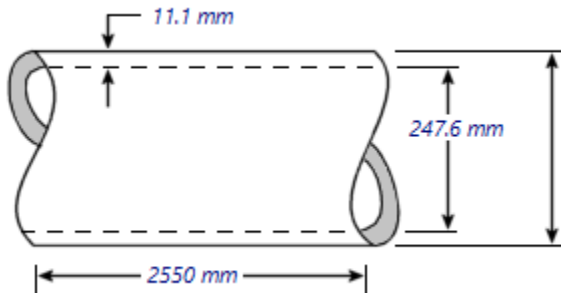
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Element: Main Shell

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	5924.995 KPa.	Int. Design Temperature	430 °C
Material	SA-335 P11	Ext. Design Temperature	430 °C
Length	2550.000 mm	External Pressure Chart	CS-2
		UNS Number	K11597
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	99.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	247.650 mm	Actual Stress	96.2 N./mm ²
Spec. Min. Thk t,e, or sa	11.1125 mm	Spec. Nominal Thk.	12.700 mm
Surface Area	21874.2 cm ²	SG of Contents	1.000
Element Volume	79236.3 cm ³	Weight of Contents	79.2 Kg
Empty Weight	205.3 Kg	Operating Weight	284.5 Kg
		Comp. All. Ope. Stress	85.9 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5925 \cdot 127) / (99 \cdot 1 - 0.6 \cdot 5925)$$

$$= 7.8750 + 3.0000 = 10.8750 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.995 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (99 \cdot 1 \cdot 8.11) / (127 + 0.6 \cdot 8.11)$$

$$= 6097 - 25 = 6072 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 11.1) / (124 + 0.6 \cdot 11.1)$$

$$= 10048 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (5925 \cdot (127 + 0.6 \cdot 8.11)) / (1 \cdot 8.11)$$

$$= 96.188 \text{ N./mm}^2$$



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 FileName : 160-C-Vertical Page 44 of 164
 Element: Main Shell Step: 9 9:37am Sep 9,2026

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 11.1, tr = 3.23, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.4, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6072.121 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	11.113 mm	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.398	Temperature Difference	23 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 40 to Tsht Ext. Chart: CS-2 at 430 °C

Elastic Modulus from Chart: CS-2 at 430 °C: 156378400 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.113	269.88	2550.00	33.27	9.4488	0.0010531	49.13

$$MAEP = (4*B)/(3*(Do/t)) = (4*49.1)/(3*33.3) = 1969 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.443	269.88	2550.00	110.47	9.4488	0.0001096	8.57

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.57)/(3*110) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.113	269.88	210101.48	33.27	50.0000	0.0010196	48.73

$$MAEP = (4*B)/(3*(Do/t)) = (4*48.7)/(3*33.3) = 1953 \text{ KPa.}$$



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Element: Top ch shell

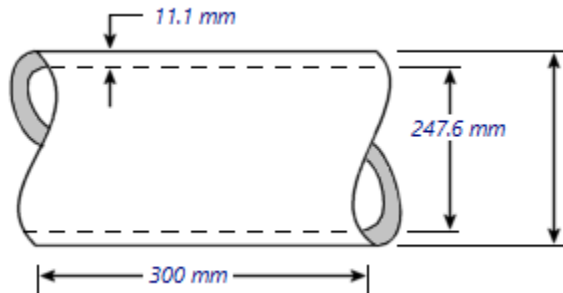
Step: 10

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top ch shell

Cylindrical Shell Design Information:			
Design Pressure P	2404.225 KPa.	Int. Design Temperature	270 °C
Material	SA-335 P11	Ext. Design Temperature	270 °C
Length	300.000 mm	External Pressure Chart	CS-2
		UNS Number	K11597
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	110.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	247.650 mm	Actual Stress	39.0 N./mm ²
Spec. Min. Thk t,e, or sa	11.1125 mm	Spec. Nominal Thk.	12.700 mm
Surface Area	2573.4 cm ²	SG of Contents	1.000
Element Volume	14453.3 cm ³	Weight of Contents	14.4 Kg
Empty Weight	24.2 Kg	Operating Weight	38.6 Kg
		Comp. All. Ope. Stress	116.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2404 \cdot 127) / (111 \cdot 1 - 0.6 \cdot 2404) \\
 &= 2.7884 + 3.0000 = 5.7884 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 4.225 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (111 \cdot 1 \cdot 8.11) / (127 + 0.6 \cdot 8.11) \\
 &= 6825 - 4.22 = 6821 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118 \cdot 1 \cdot 11.1) / (124 + 0.6 \cdot 11.1) \\
 &= 10048 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2404 \cdot (127 + 0.6 \cdot 8.11)) / (1 \cdot 8.11) \\
 &= 39.031 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 160-C-Vertical Page 46 of 164
 Element: Top ch shell Step: 10 9:37am Sep 9,2026

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 11.1, tr = 2.01, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.25, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3764.250 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	11.113 mm	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.248	Temperature Difference	78 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Top ch shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 270 °C

Elastic Modulus from Chart: CS-2 at 270 °C: 184606480 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.113	269.88	300.03	33.27	1.1117	0.0064663	94.58

$$MAEP = (4*B)/(3*(Do/t)) = (4*94.6)/(3*33.3) = 3791 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
0.935	269.88	300.03	288.72	1.1117	0.0002426	22.40

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.4)/(3*289) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.113	269.88	9395.75	33.27	34.8152	0.0010199	67.78

$$MAEP = (4*B)/(3*(Do/t)) = (4*67.8)/(3*33.3) = 2717 \text{ KPa.}$$



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Shop Pressure Test Results:

Step: 11

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7893.756 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7670.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13058.051 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	6679.333 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8437.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8437.000 KPa.
App 27-4	= M.A.W.P.	6072.121 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4893.525 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3120.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6449.185 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	4140.676 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3432.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3432.000 KPa.
App 27-4	= M.A.W.P.	3764.250 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Bottom ch shell	75.8	184.5	0.411	6451.61
Main Shell	153.4	184.5	0.831	13060.48
Top ch shell	75.8	184.5	0.411	6451.61

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	110.80	1.065
N3-Shell inlet	Nozzle	118.00	98.98	1.192
N3-Shell inlet	Pad	138.00	80.48	1.715
N4-Shell outlet	Nozzle	118.00	98.98	1.192
N2-Tube Outlet	Nozzle	118.00	110.80	1.065
N2-Tube Outlet	Pad	138.00	137.20	1.006

Minimum

1.006

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Blind flange	138.00	138.00	1.000
Bottom ch flg	138.00	138.00	1.000
Bottom ch shell	118.00	110.80	1.065
Main Shell	118.00	98.98	1.192
Top ch shell	118.00	110.80	1.065
Top ch flg	118.00	110.80	1.065
Top Blind Flang	118.00	110.80	1.065

Minimum

1.000



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Shop Pressure Test Results:

Step: 11

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Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	110.80	1.065
Tubesheet Material	148.00	148.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	14.99	184.50	0.081
N3-Shell inlet	61.87	184.50	0.335
N4-Shell outlet	27.63	184.50	0.150
N2-Tube Outlet	24.76	184.50	0.134

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 12 9:37am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7893.756 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7670.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13037.271 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	6679.333 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8437.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8437.000 KPa.
App 27-4	= M.A.W.P.	6072.121 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4893.525 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3120.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6452.959 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	4140.676 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3432.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3432.000 KPa.
App 27-4	= M.A.W.P.	3764.250 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 5900 • 1

= 7670.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 2400 • 1

= 3120.000 KPa.

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Bottom ch shell	51.2	184.5	0.277	3153.07
Main Shell	124.9	184.5	0.677	7695.00
Top ch shell	50.7	184.5	0.275	3124.16

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	110.80	1.065
N3-Shell inlet	Nozzle	118.00	98.98	1.192
N3-Shell inlet	Pad	138.00	80.48	1.715
N4-Shell outlet	Nozzle	118.00	98.98	1.192
N2-Tube Outlet	Nozzle	118.00	110.80	1.065
N2-Tube Outlet	Pad	138.00	137.20	1.006

Minimum

1.006

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
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Field Pressure Test Results: Step: 12 9:37am Sep 9, 2026

Blind flange	138.00	138.00	1.000
Bottom ch flg	138.00	138.00	1.000
Bottom ch shell	118.00	110.80	1.065
Main Shell	118.00	98.98	1.192
Top ch shell	118.00	110.80	1.065
Top ch flg	118.00	110.80	1.065
Top Blind Flang	118.00	110.80	1.065

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	110.80	1.065
Tubesheet Material	148.00	148.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	7.33	184.50	0.040
N3-Shell inlet	36.45	184.50	0.198
N4-Shell outlet	16.28	184.50	0.088
N2-Tube Outlet	11.99	184.50	0.065

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 9:37a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		7670.00 KPa.
Shell Thickness	ts	12.7000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	247.650 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-335 P11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11597
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: Bottom ch shell

Channel Type:		Cylinder
Channel Test Pressure		3120.00 KPa.
Channel Thickness	tc	12.7000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	247.650 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-335 P11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11597
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	60
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	2650.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2550.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	101.600 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	268.2 cm ²



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ASME TS Calc: Pressure Test

Case: 1 9:37a Sep 9,2026

Perimeter of Tube Layout Cp 609.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K11789
Tubesheet Allowable Stress at Temperature S 148.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 148.00 N./mm²
Thickness of Tubesheet h 50.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 269.278 mm

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 370.000 mm
Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Shell Side Min. Design Pressure Psd,min -103.4190 KPa.
Channel Side Min. Design Pressure Ptd,min -103.4190 KPa.

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



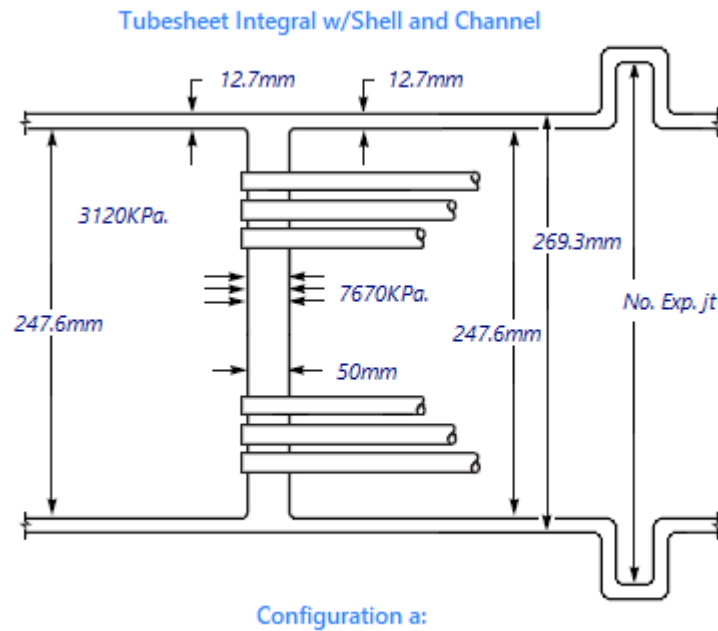
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ASME TS Calc: Pressure Test

Case: 1 9:37a Sep 9, 2026



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Note:



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ASME TS Calc: Pressure Test

Case: 1 9:37a Sep 9, 2026

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (3223 \cdot (19/2 - 0)) / (118 \cdot 1 + 0.4 \cdot 3223)$$

$$= 0.2574 + 0.0000 = 0.2574 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 21.11 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	2550.00	9.03	50.0000	0.0134948	122.73
EMAWP = (2.167 / (D/T) - 0.0833) * B = 19233 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.9932	19.05	2550.00	19.18	50.0000	0.0029898	111.85
EMAWP = (4 * B) / (3 * (D/T)) = (4 * 112) / (3 * 19.2) = 7774 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.9932 mm
Allowable Internal Pressure at Corroded thickness	28678.85 KPa.
Required Internal Design Pressure	3223.42 KPa.
Allowable External Pressure at Corroded thickness	19233.22 KPa.
Required External Design Pressure	7773.42 KPa.
Required Thickness due to Shell Side pressure	0.9932 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/50$$

$$= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$$

$$= \max(19 - 2 \cdot 2.11 (204080160/204080160) \cdot (118/148) \cdot (0); 19 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 102 + 19$$

$$= 222.250 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (25.4 - 19) / 25.4$$

$$= 0.250$$



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Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= D_o / 2 \\ &= 222 / 2 \\ &= 111.1250 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) } \\ &= 25.4 / \sqrt{1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 222 \cdot 25.4) / (\pi \cdot 222^2) } \\ &= 25.4000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (25.4 - 19) / 25.4 \\ &= 0.2500 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= a_s / a_o \\ &= 124 / 111 \\ &= 1.1143 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= a_c / a_o \\ &= 124 / 111 \\ &= 1.1143 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2 \\ &= 1 - 60 ((19 - 2 \cdot 2.11) / (2 \cdot 111))^2 \\ &= 0.7329 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot a_o))^2 \\ &= 1 - 60 (19 / (2 \cdot 111))^2 \\ &= 0.5592 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned} &= \max((h_g - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((0 - 0); 0) \\ &= 0.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$\begin{aligned} &= \pi \cdot t_s(D_s + t_s) E_s / L \\ &= \pi \cdot 12.7 (248 + 12.7) 204080160 / 2550 \\ &= 84775.0938 \text{ Kg/mm} \end{aligned}$$

Axial Tube Stiffness [Kt]:

$$\begin{aligned} &= \pi \cdot t_t(D_t - t_t) E_t / L \\ &= \pi \cdot 2.11 (19 - 2.11) 204080160 / 2550 \\ &= 916.4375 \text{ Kg/mm} \end{aligned}$$

Stiffness Factor [Ks,t]:

$$\begin{aligned} &= K_s / (Nt \cdot K_t) \\ &= 84775 / (60 \cdot 916) \\ &= 1.54175 \end{aligned}$$

Rigidity Ratio [J]:

$$\begin{aligned} &= 1 / (1 + K_s / K_j) \\ &= 1 / (1 + 84775 / 0) \\ &= 1.0000 (= 1 \text{ if No Exp. Jt.}) \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s) \\ &= ((12(1 - 0.3^2))^{1/4}) / ((248 + 12.7) 12.7) \end{aligned}$$



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= 0.0316 1/mm

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.032 \cdot 204080160 \cdot 12.7^3 / (6(1 - 0.3^2))$$

$$= 2420439808.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(248 + 12.7) \cdot 2.42E+09 / 50^3 \cdot (1 + 50 \cdot 0.032 + (50 \cdot 0.032)^2 / 2)$$

$$= 115847800.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 248^2 / (4 \cdot 204080160 \cdot 12.7) \cdot (1 - 248 \cdot 0.3 / (2(248 + 12.7)))$$

$$= 0.50714E-02 \text{ mm/N./mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(248 + 12.7)12.7}$$

$$= 0.0316 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.032 \cdot 204080160 \cdot 12.7^3 / (6(1 - 0.3^2))$$

$$= 2420439808.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(248 + 12.7 / \cos(0)) \cdot 2.42E+09 / 50^3 \cdot (1 + 50 \cdot 0.032 \cdot (1 + 1) / 2 + (50 \cdot 0.032)^2 / 2 \cdot 1)$$

$$= 115847800.0 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 248^2 / (4 \cdot 204080160 \cdot 12.7) \cdot (1 - 248 \cdot 0.3 / (2(248 + 12.7)))$$

$$= 0.50714E-02 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

h/p = 1.968504 ; μ* = 0.250000 ; E* = 44422412 KPa.

$$E^*/E = 0.217671 ; \nu^* = 0.395112$$

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t)_{ao^2}) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.4^2)60 \cdot 204080160 \cdot 2.11(19 - 2.11)^{111^2}) / (44422412 \cdot 2550 \cdot 50^3))^{1/4}$$

$$= 1.6675$$

Values from Table 4.18.3

$$Z_d = 0.309596 ; Z_v = 0.166405 ; Z_m = 0.678824$$

$$Z_a = 0.638921E+00 ; Z_w = 0.166405$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 269/222$$



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= 1.2116

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.4) / 44422412 \cdot (115847800 + 115847800 + 204080160 \cdot \ln(1.21))$$

= 3.6883

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.4) 3.69$$

= 5.1456

Compute parameter [Q1]:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.11 - 1 - 5.15 \cdot 0.17) / (1 + 5.15 \cdot 0.68)$$

= -0.165140

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.31 + -0.17 \cdot 0.17) / 2 \cdot 1.67^4$$

= 1.0905

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.17 + -0.17 \cdot 0.68) / 2 \cdot 1.67^4$$

= 0.2099

Compute parameter [U]:

$$= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.17 + (1.11 - 1) 0.68) 1.67^4 / (1 + 5.15 \cdot 0.68)$$

= 0.4198

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

= 0 (config a)

Compute Parameter [γ]:

= 0.000 mm (For Pressure only cases)

Calculate parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.11 \cdot 2.42E+09 \cdot 0.032 \cdot 0.005071407 (1 + 50 \cdot 0.032)$$

= 1115.9744 mm²

Calculate parameter [ω^*s]:

$$= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 111^2 (1.11^2 - 1) \cdot (1.11 - 1) / 4 - 1116$$

= -1030.7212 mm²

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.11 \cdot 2.42E+09 [0.005071407 (1 + 0.032 \cdot 1 \cdot 50) - 0.000 (1/50 + 1 \cdot 0.032/2)]$$

= 1115.9744 mm²

Parameter [ω^*c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 111^2 [(1.11^2 + 1) \cdot (1.11 - 1) / 4 - (1.11 - 1) / 2] - 1116$$

= -1030.7212 mm²

Pressure [P^*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D



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Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 7670 \{ 0.56 + 2(1 - 0.56) 0.3 + [2/1.54 \cdot (248/222)^2] 0.3 - [(1.11^2 - 1)/(1 \cdot 1.54)] - [(1 - 1)/(2 \cdot 1 \cdot 1.54)] [(0^2 - (248)^2)/222^2] \}$$

$$= 8821.6582 \text{ KPa.}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.73 + 2(1 - 0.73) 0.3 + 1/(1 \cdot 1.54)] 3120$$

$$= 4810.2734 \text{ KPa.}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 60 \cdot 916 \cdot 0 / (3.14 \cdot 111^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 0.42 \cdot 0 / (2 \cdot \pi \cdot 111^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_t)$$

$$= -(0.42/111^2) \cdot (-1031 \cdot 7670 - -1031 \cdot 3120)$$

$$= 159.4306 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_c)$$

$$= (0.42/111^2) \cdot (1116 \cdot 0 - 1116 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (p_s - 1)Qz2)) \cdot (P_s - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 1.54 / (1 + 1 \cdot 1.54(1.09 + (1.11 - 1)0.21)) \cdot (8822 - 4810 + 0 + 0 + -0 + 159)$$

$$= 2365.6582 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega_s \cdot P_s - \omega_c \cdot P_t) - (\omega_s \cdot P_s - \omega_c \cdot P_c)) + W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-1031 \cdot 7670 - -1031 \cdot 3120) - (1116 \cdot 0 - 1116 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 5.15 \cdot 0.68)$$

$$= -106.4425 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2)$$

$$= -0.17 + 2 \cdot -106 / (2366 \cdot 111^2)$$

$$= -0.23660$$

Fm Value from Table 4.18.3 = 0.118301

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe$$



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$$= (1.5 \cdot 0.12 / 0.25) \cdot (2 \cdot 111 / (50 - 0))^2 \cdot 2366$$

$$= 33.1787 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 310 = 442 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.044 / 0.25) \cdot (2 \cdot 111 / (5.7 - 0))^2 \cdot 1090$$

$$= 441.7314 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 5.7 + 0 + 0$$

$$= 5.6995 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$$

$$= (2 \cdot 0.25 \cdot 50 / 111) \cdot 236$$

$$= 53.0024 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

Tubesheet thickness (Incl. Corr.) = 3.8100 mm

Tubesheet Shear Stress = 40.3020 N./mm²

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(5.7; 3.81)$$

$$= 5.6995 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.92, F_{tmax} = 1.04$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((7670 \cdot 0.56 - 3120 \cdot 0.73) - (2366) \cdot 0.92) / (0.73 - 0.56)$$

$$= -0.9635 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((7670 \cdot 0.56 - 3120 \cdot 0.73) - (2366) \cdot 1.04) / (0.73 - 0.56)$$

$$= -2.6978 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 2.7 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta T \cdot S_{y,t} = 194.7465 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.31 + -0.24 \cdot 0.17)1.67^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2)$$



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Determine the factor [rt]:

$$= \left(\sqrt{\left(\frac{dt^2}{4} + \left(\frac{dt}{2} - 2 \cdot \frac{tt^2}{4} \right)^2} \right) \right) / 4$$

$$= \left(\sqrt{\left(\frac{19^2}{4} + \left(\frac{19}{2} - 2 \cdot \frac{2.11^2}{4} \right)^2} \right) \right) / 4$$

$$= 6.0355 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{\left(\frac{2 \cdot \pi^2 \cdot Et}{Syt} \right)}$$

$$= \sqrt{\left(\frac{2 \cdot \pi^2 \cdot 204080160}{205} \right)}$$

$$= 149.3679$$

Determine the factor [Ft]:

$$= k \cdot L/r$$

$$= 1 \cdot 370/6.04$$

$$= 61.3042$$

Buckling Allowable Stress [Stb]:

$$= (Sy, t/Fs) \cdot (1 - Ft/(2 \cdot Ct))$$

$$= (205/2) \cdot (1 - 61.3/(2 \cdot 149))$$

$$= 194.746 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t, \max} \cdot \text{Tube Area}$$

$$= 2.7 \cdot 1.12$$

$$= 30.89 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 2.11) 2.11$$

$$= 1.123 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19 - 2.11) 195$$

$$= 2229.907 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 195$$

$$= 2229.9067 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 195/195$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 195$$

$$= 1481.5632 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{\left(\frac{0.75 \cdot d_o}{2} \right)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{\left(\frac{0.75 \cdot 19}{2} \right)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) 1 \cdot 1} - 0.75 \cdot 19 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm



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Tube-Tubesheet Jt allowable, 2230 is >= tube strength 2230 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 205 = 292 \text{ N./mm}^2$$

Axial Membrane Stress in Shell $[\sigma_s, m]$:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 111^2 / ((248 + 12.7) 12.7) \cdot [2366 + (1.11^2 - 1)(7670 - 3120)] + 124^2 \cdot 3120 / ((248 + 12.7) 12.7)$$

$$= 27.4109 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 12.7 mm, the minimum Shell length adjacent to the tubesheet should be 100.947 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	33.2 <=	441.7 N./mm ²	Pass
Tubesheet Shear Stress	8.3 <=	235.6 N./mm ²	Pass
Maximum Tube Stress	2.7 <=	194.7 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-2.7 <=	-194.7 N./mm ²	Pass
Maximum Force on any one Tube	30.9 <=	2229.9 Kgf	Pass
Axial Membrane Stress in Shell	27.4 <=	292.1 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	5.700	50.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	7.131 ...	35	442	10 236	Ps+Pt-Th	D1 Ok
D2uc	14.222 ...	68	442	18 236	Ps+Pt-Th	D2 Ok
D3uc	5.700 ...	33	442	8 236	Ps+Pt-Th	D3 Ok
Max:	14.2224 ... mm	0.155	0.078 (Str. Ratio)			

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx



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[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:		Fail
D1uc	1	292	...	...	:	...	...	...	...	:		Ok
D2uc	26	292	...	...	:	...	...	...	...	:		Ok
D3uc	27	292	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.094		...		:	...		...		:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	4	194	...	...	46	2229	...	292	...	292	Ok
D2uc	7	194	-7	-194	77	2229	...	292	...	292	Ok
D3uc	3	194	-3	-194	31	2229	...	292	...	292	Ok
Max RATIO	0.035		0.035		0.035		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	14.222	50.000	Ok
Tubesheet thickness flanged extension :	0.000	14.000	Ok
Tube thickness :	0.993	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (12.700) adj. to tubesheet: 100.947 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.114 mm
 Due to Pressure + Thermal Shell Compresses by : -0.114 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0146	...	...	...	...	...
2	-0.1139	...	...	...	...	...
3	-0.1004	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...



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ASME TS Calc: Pressure Test

Case: 1 9:37a Sep 9, 2026

6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	42	No Calc	No Calc
20	30	No Calc	...	52	No Calc	No Calc
30	Tsht	299.969	269.875	8.1125	0.0064678	94.5862
Tsht	40	No Calc	...	8.1125	No Calc	No Calc
40	Tsht	2550	269.875	8.1125	0.0010531	49.1306
Tsht	50	No Calc	...	8.1125	No Calc	No Calc
50	60	300.031	269.875	8.1125	0.0064663	94.583
60	70	No Calc	...	52	No Calc	No Calc
70	80	No Calc	...	42	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	45	37.0586	103.419	No Calc
20	30	55	46.736	103.419	No Calc
30	Tsht	11.1125	3.93467	103.419	3791
Tsht	40	11.1125	No Calc	103.419	No Calc
40	Tsht	11.1125	5.44306	103.419	1969
Tsht	50	11.1125	No Calc	103.419	No Calc
50	60	11.1125	3.93474	103.419	3791
60	70	55	51.181	103.419	No Calc
70	80	45	39.8272	103.419	No Calc

Minimum

1969

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	299.969	9394.02	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	2550	210101	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	300.031	9395.75	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 14

9:37am

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	59.653	...	59.653	...	11.9306
20	30	48.8666	4289.76	47.292	4346.46	9.77331
30	40	24.1525	14453.3	18.6598	15162.1	4.8305
40	50	205.296	79236.3	158.608	85261.3	41.0593
50	60	24.1525	14453.3	18.6598	15162.1	4.8305
60	70	48.8666	4289.76	47.292	4346.46	9.77331
70	80	59.653	...	59.653	...	11.9306
Total		470	116722.35	409	124278.41	94



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	...	...	25	...	Liquid: 10
10	Insl	9.44105	...	25	...	Ins: 70
20	Liqd	4.14074	...	43	...	Liquid: 20
20	Insl	7.33068	...	43	...	Ins: 70
30	Liqd	14.4444	...	150	...	Liquid: 30
30	Insl	12.2851	...	150	...	Ins: 70
30	Nozl	8.57494	...	150	198.831	N1-Tube Inlet
40	Liqd	79.188	...	1275	...	Liquid: 40
40	Insl	177.808	...	1275	...	Ins: 70
40	Nozl	37.1279	...	500	222.25	N3-Shell inlet
40	Nozl	12.7876	...	2050	-222.25	N4-Shell outlet
40	Lugs	105.785	...	1426	...	Support Lugs
50	Liqd	14.4444	...	150	...	Liquid: 50
50	Insl	12.2851	...	150	...	Ins: 70
50	Nozl	32.6638	...	150	-198.831	N2-Tube Outlet
60	Liqd	4.14074	...	43	...	Liquid: 60
60	Insl	7.33068	...	43	...	Ins: 70
70	Liqd	...	...	22.5	...	Liquid: 70
70	Insl	9.44105	...	25	...	Ins: 70
30	FTsh	18.5307	...	325	...	Fixed Tubesheet
30	Tube	166.056	...	1625	...	Tubes
30	RTsh	18.5307	...	2925	...	Tubesheet
40	Bafl	4.04035	...	370	...	Baffles
40	Bafl	4.04035	...	740	...	Baffles
40	Bafl	4.04035	...	1110	...	Baffles
40	Bafl	4.04035	...	1480	...	Baffles
40	Bafl	4.04035	...	1850	...	Baffles
40	Bafl	4.04035	...	2220	...	Baffles
40	Bafl	4.10386	...	2590	...	Baffles

Total Weight of Each Detail Type:

Liquid	116.4
Insulation	235.9
Nozzles	91.2
Lugs	105.8



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Element and Detail Weights:

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Exchanger Components	231.5
Liquid in Tubes	27.5

Sum of the Detail Weights 808.1 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	564.8	564.8	564.8	564.8	564.8	564.8
Nozzles	91.2	91.2	91.2	91.2	91.2	91.2
Lugs	105.8	105.8	105.8	105.8	105.8	105.8
Exchanger	231.5	231.5	231.5	231.5	231.5	231.5
Insulation	...	...	...	...	235.9	235.9 *
Ope. Liquid	...	...	...	...	...	116.4
Tube Ope Lqd	...	...	...	...	...	27.5
Test Liquid	...	116.7	...	...	...	...
Tube Tst Lqd	...	27.5	...	...	...	...
Totals	993.2	1137.3	993.2	993.2	1229.1	1372.9

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 20.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	993.2 kg.
Shop Test Wt.	- Fabricated weight + water (full)	1137.3 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	993.2 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	993.2 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	1229.1 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	1372.9 kg.
Field Test Wt.	- Empty weight + test medium (full)	1373.2 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	365.0 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	323.7 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	348.6 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	27469.3 cm ³
Weight of Ope Liq in tubes :	27.5 kg.
Weight of Water in tubes :	27.5 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	2168.17	...
20	30	2021.93	669.093
30	40	2573.44	2334.05
40	50	21874.2	19839.4
50	60	2573.44	2334.05
60	70	2021.93	669.093
70	80	2168.17	...



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Element and Detail Weights:

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Total	35401.3	25845.7 cm ²
Total	38.1	27.8 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	71.5836	81.0247	81.0247	...	81.0247
20	30	58.6399	70.1113	70.2577	...	65.9705
30	40	56.0887	82.8181	82.8181	1.705	68.3737
40	Lugs	328.492	484.543	484.543	5.40974	417.466
Lugs	50	162.181	290.579	290.579	...	251.016
50	60	80.1775	106.907	106.907	6.4947	92.4625
60	70	58.6399	70.1113	70.2577	...	65.9705
70	80	71.5836	81.0247	81.0247	...	81.0247

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	...	...	...
20	30	-81.0247	-81.0247	-81.0247
30	40	-146.995	-151.136	-151.282
40	Lugs	-215.369	-233.954	-234.101
Lugs	50	490.473	548.622	548.768
50	60	239.458	258.043	258.189
60	70	146.995	151.136	151.282
70	80	81.0247	81.0247	81.0247

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	...	...	...
20	30	...	...	...
30	40	1.705	1.705	1.705
40	Lugs	7.11474	7.11474	7.11474
Lugs	50	6.4947	6.4947	6.4947
50	60	6.4947	6.4947	6.4947
60	70	...	...	...
70	80	...	...	...



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Nozzle Flange MAWP:

Step: 15 9:37am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	UG-44	Max Pressure 50%	DNV
N1-Tube Inlet	8226.1	10200.0	270	600	GR 1.1	8574.9	3764.3	8226.1	3764.2
N3-Shell inlet	8274.7	15310.0	430	900	GR 1.1	634.6	8274.7	8274.7	8274.7
N4-Shell outlet	8274.7	15310.0	430	900	GR 1.1	1106.8	8274.7	8274.7	8274.7
N2-Tube Outlet	8226.1	10200.0	270	600	GR 1.1	6481.2	5857.9	8226.1	5857.9

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 8274.691 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 15310.003 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 3764.250 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 10200.003 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Tube Inlet	77.851	0.50	72	459	3764.3	3764.3	Pass
N3-Shell inlet	135.001	0.50	0	918	6072.1	8274.7	Pass
N4-Shell outlet	72.263	0.50	0	459	6072.1	8274.7	Pass
N2-Tube Outlet	135.001	0.50	288	918	3764.3	5857.9	Pass

Evaluating Flange on Nozzle [N1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(72) + 4 \cdot \text{abs}(459)77.851 \leq \pi \cdot 77.851^3 [(8226.086 - 3764.250) + 500 \cdot 8226.086]$$

$$= 0.12962E+04 \leq 0.12962E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8226.086, 0.500 \cdot 8226.086 + 8226.086 - 1296.2 / (\pi \cdot 77.851^3))$$

$$= \min(8226.086, 3764.250)$$

$$= 3764.250 \text{ KPa.}$$

Evaluating Flange on Nozzle [N3-Shell inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(0) + 4 \cdot \text{abs}(918)135.001 \leq \pi \cdot 135.001^3 [(8274.691 - 6072.121) + 500 \cdot 8274.691]$$

$$= 0.50024E+03 \leq 0.49975E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Shell inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8274.691, 0.500 \cdot 8274.691 + 8274.691 - 500.2 / (\pi \cdot 135.001^3))$$

$$= \min(8274.691, 11777.417)$$

$$= 8274.691 \text{ KPa.}$$

Evaluating Flange on Nozzle [N4-Shell outlet]:



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Nozzle Flange MAWP:

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$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(0) + 4 \cdot \text{abs}(459) 72.263 \leq \pi \cdot 72.263^3 [(8274.691 - 6072.121) + .500 \cdot 8274.691] \\
 &= 0.13380E+03 \leq 0.76645E+03 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4-Shell outlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(8274.691, 0.500 \cdot 8274.691 + 8274.691 - 133.8 / (\pi \cdot 72.263^3)) \\
 &= \min(8274.691, 11305.260) \\
 &= 8274.691 \text{ KPa.}
 \end{aligned}$$

Evaluating Flange on Nozzle [N2-Tube Outlet]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(288) + 4 \cdot \text{abs}(918) 135.001 \leq \pi \cdot 135.001^3 [(8226.086 - 3764.250) + .500 \cdot 8226.086] \\
 &= 0.51089E+04 \leq 0.67592E+04 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N2-Tube Outlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(8226.086, 0.500 \cdot 8226.086 + 8226.086 - 5108.9 / (\pi \cdot 135.001^3)) \\
 &= \min(8226.086, 5857.879) \\
 &= 5857.879 \text{ KPa.}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Natural Frequency Calculation: Step: 16 9:37am Sep 9, 2026

The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 24.7663 Hz.

The Natural Frequency for the Vessel (Ope...) is 23.5442 Hz.

The Natural Frequency for the Vessel (Filled) is 25.2689 Hz.

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Wind Load Calculation:

Step: 17 9:37am Sep 9, 2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	92.7 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.97) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$$

$$= \min(0.85, 0.91)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	23.544 Hz
Natural Frequency of Vessel (Empty)	24.766 Hz
Natural Frequency of Vessel (Test)	25.269 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.593
Structure Height to Diameter ratio	6.593
Height to top of Structure	3504.350 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm}$

Effective Height $[z]$:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 25.7 + 300 = 326 \text{ mm}$$

$$= 1.07 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z $[Kz]$:

$$\text{Because } z (1.07 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.01(15 / Zg)^{2/\alpha}$$

$$= 2.01(15/900)^{2/9.5}$$

$$= 0.85$$

Type of Hill: No Hill



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Wind Load Calculation:

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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 57.6²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.59 • 0.37

= 2.95 lbs. [1.34] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Blind flange	325.7	0.000	0.000	0.000	0.849	1.000	78.120
Bottom ch flg	387.7	0.000	0.000	0.000	0.849	1.000	78.120
Bottom ch shell	577.2	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2052.2	0.000	0.000	0.000	0.849	1.000	78.120
Top ch shell	3527.2	0.000	0.000	0.000	0.849	1.000	78.120
Top ch flg	3719.9	0.000	0.000	0.000	0.849	1.000	78.120
Top Blind Flang	3781.8	0.000	0.000	0.000	0.849	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of (D/L²)(104)
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.32037E-06 (High); 0.32037E-06 < 0.40047E-06 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: 0/3504 = 0

:: [D / L²] • 10⁴ < 8 (English Units, ft.)

:: [1.74/11.5²] • 10⁴ = 132 [Geometry Violation]



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Wind Load Calculation:

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Compute the vibration possibility. If $V_p > 0.40047E-06$ no chance. $[V_p]$:

$$= W / (L \cdot Dr^2)$$

$$= 1300 / (3504 \cdot 464^2)$$

$$= 0.17234E-05$$

Since V_p is $> 0.40047E-06$ no further vibration analysis is required !

The Natural Frequency for the Vessel (Ope...) is 23.5442 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	325.675	756	340.2	78.12	1.4741
20	30	387.675	537.18	424.372	78.12	1.83882
30	40	577.175	563.85	1691.55	78.12	7.32956
40	50	2052.18	683.85	17438.2	78.12	75.5604
50	60	3527.17	563.85	1691.55	78.12	7.32956
60	70	3719.85	537.18	424.372	78.12	1.83882
70	80	3781.85	756	340.2	78.12	1.4741

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation:

Step: 18

9:37am

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		2.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	2.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\
 SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_n^x$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(11.50.75)$
 $= 0.125$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/23.5)$
 $= 0.042$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.12, 0.042)$ per 12.8.2
 $= 0.042$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 1267 \cdot 1.25$
 $= 74.507$ Kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 1267$
 $= 39.7$ Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1



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Earthquake Load Calculation:

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The Natural Frequency for the Vessel (Ope...) is 23.5442 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	25.675	81.0247	0.064105	0.057145	-2.54093
20	30	87.675	70.1113	0.18942	0.15985	-4.73962
30	40	277.175	82.8181	0.70736	0.48337	-7.3368
40	Lugs	1703.18	484.543	25.4304	17.3955	-22.5321
Lugs	50	2390.18	290.579	21.402	12.0527	17.2048
50	60	3227.17	106.907	10.6314	8.04502	8.09223
60	70	3419.85	70.1113	7.3885	6.23523	4.73962
70	80	3481.85	81.0247	8.69339	7.74955	2.54093

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Wind/Earthquake Shear, Bending: Step: 19 9:37am Sep 9,2026

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1727.5	...	...	...	...
20	30	1665.5	1.4741	0.064105	0.033168	0.0014424
30	40	1476	3.31293	0.25352	0.22226	0.013989
40	Lugs	638	10.6425	0.96088	2.31561	0.19615
Lugs	50	637	96.8454	74.5066	0.086332	38.9519
50	60	1474	10.6425	26.7133	2.32613	7.64471
60	70	1666.68	3.31293	16.0819	0.22226	1.17425
70	80	1728.67	1.4741	8.69339	0.033168	0.1956

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind/Earthquake Deflection:

Step: 20

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	0.021967	0.023206	0.97085E-06
20	30	1.4741	0.018555	0.020729	0.97078E-06
30	40	3.31293	0.0081535	0.016381	0.96852E-06
40	Lugs	10.6425	-0.030843	...	...
Lugs	50	96.8454	0.001272	0.056016	0.20086E-04
50	60	10.6425	0.069243	0.082491	0.19790E-04
60	70	3.31293	0.085981	0.089471	0.19789E-04
70	80	1.4741	0.091459	0.093447	0.19789E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	319.526	1997.04
20	30	227.041	1419.01
30	40	238.313	1489.46
40	50	289.032	1806.45
50	60	238.313	1489.46
60	70	227.041	1419.01
70	80	319.526	1997.04

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)

Allowable deflection : 17.522 Actual deflection : 0.093 mm

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Longitudinal Stress Constants: Step: 21 9:37am Sep 9,2026

Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	522.942	499.319	3458814	3355056
20	30	90.3364	66.7133	561362	423859
30	40	90.3364	66.7133	561362	423859
40	50	90.3364	66.7133	561362	423859
50	60	90.3364	66.7133	561362	423859
60	70	90.3364	66.7133	561362	423859
70	80	90.3364	66.7133	561362	423859

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Longitudinal Allowable Stresses: Step: 22 9:37am Sep 9,2026

Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	165.597	296.995	-138.355	-147.277
20	30	165.597	296.995	-116.044	-147.277
30	40	132.959	221.396	-116.044	-147.277
40	Lugs	118.778	221.396	-85.9199	-147.277
Lugs	50	118.778	221.396	-85.9199	-147.277
50	60	132.959	221.396	-116.044	-147.277
60	70	132.959	221.396	-116.044	-147.277
70	80	132.959	221.396	-116.044	-147.277

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Longitudinal Stresses due to:

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	18.281	-0.88681	23.7653
40	50	44.9408	-0.88681	58.423
50	60	18.281	-0.88681	23.7653
60	70	...	...	...
70	80	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	...	...	...	...	...
20	30	0.11911	0.11911	0.11911	...	...
30	40	0.21608	0.22217	0.22238	0.039448	0.039448
40	Lugs	0.31659	0.34391	0.34413	0.16461	0.16461
Lugs	50	-0.72099	-0.66275	-0.72099	0.15027	0.15027
50	60	-0.352	-0.352	-0.352	0.15027	0.15027
60	70	-0.21608	-0.21608	-0.21608	...	...
70	80	-0.11911	-0.11911	-0.11911	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	...	...	...	...
20	30	...	0.00076739	0.33372E-04	0.0003837	...
30	40	0.039448	0.0051423	0.00032366	0.0025712	...
40	Lugs	0.16461	0.053575	0.0045383	0.026788	...
Lugs	50	0.15027	0.0019974	0.90121	0.00099872	...
50	60	0.15027	0.053819	0.17687	0.026909	...
60	70	...	0.0051423	0.027168	0.0025712	...
70	80	...	0.00076739	0.0045256	0.00038369	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	...
20	30	...	...	...	0.29749E-04
30	40	...	...	...	0.00028029



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Longitudinal Stresses due to:

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40	Lugs	...	...	...	0.0034641
Lugs	50	...	...	...	0.72067
50	60	...	...	...	0.14992
60	70	...	...	...	0.023898
70	80	...	...	...	0.0040342

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	-0.00049905
20	30	...	-0.0069672
30	40	...	-0.010785
40	Lugs	...	-0.033122
Lugs	50	...	0.025291
50	60	...	0.011896
60	70	...	0.0069672
70	80	...	0.0037352

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	Lugs	...	...	...	...
Lugs	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	Lugs	...
Lugs	50	...
50	60	...
60	70	...
70	80	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	0.26	132.96		116.04	0.0020	
40	0.53	118.78		85.92	0.0045	
40		118.78	-0.87	85.92		0.0102
50		132.96	-0.56	116.04		0.0048

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	0.25	132.96		116.04	0.0018	
40	0.45	118.78		85.92	0.0038	
40	0.18	118.78	-1.57	85.92	0.0015	0.0182
50		132.96	-0.64	116.04		0.0055

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	0.27	132.96		116.04	0.0020	



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40	0.56	118.78		85.92	0.0047	
40		118.78	-0.82	85.92		0.0095
50		132.96	-0.56	116.04		0.0048

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	0.25	132.96		116.04	0.0019	
40	0.48	118.78		85.92	0.0040	
40	0.41	118.78	-1.69	85.92	0.0035	0.0197
50		132.96	-0.67	116.04		0.0057

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	0.26	221.40		147.28	0.0012	
40	0.54	221.40		147.28	0.0024	
40		221.40	-0.87	147.28		0.0059
50		221.40	-0.53	147.28		0.0036

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	0.26	221.40		147.28	0.0012	
40	0.51	221.40		147.28	0.0023	
40		221.40	-0.87	147.28		0.0059
50		221.40	-0.50	147.28		0.0034

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	18.55	132.96		116.04	0.1395	
40	45.50	118.78		85.92	0.3831	
40	44.43	118.78		85.92	0.3741	
50	18.13	132.96		116.04	0.1364	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	18.53	132.96		116.04	0.1394	
40	45.42	118.78		85.92	0.3824	
40	45.35	118.78		85.92	0.3818	
50	18.27	132.96		116.04	0.1374	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30		132.96	-0.71	116.04		0.0061
40		118.78	-0.76	85.92		0.0089
40		118.78	-1.70	85.92		0.0198
50		132.96	-1.44	116.04		0.0124

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30		132.96	-0.72	116.04		0.0062
40		118.78	-0.75	85.92		0.0087
40		118.78	-2.58	85.92		0.0300



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Stress due to Combined Loads: Step: 24 9:37am Sep 9,2026

50	132.96	-1.55	116.04	0.0134
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Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	24.03	221.40		147.28	0.1085	
40	58.96	221.40		147.28	0.2663	
40	57.85	221.40		147.28	0.2613	
50	23.59	221.40		147.28	0.1066	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	24.03	221.40		147.28	0.1085	
40	58.93	221.40		147.28	0.2662	
40	57.85	221.40		147.28	0.2613	
50	23.56	221.40		147.28	0.1064	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	18.54	132.96		116.04	0.1394	
40	45.42	118.78		85.92	0.3824	
40	44.37	118.78		85.92	0.3736	
50	18.08	132.96		116.04	0.1360	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	18.45	132.96		116.04	0.1387	
40	45.19	118.78		85.92	0.3805	
40	44.45	118.78		85.92	0.3742	
50	18.02	132.96		116.04	0.1355	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	18.54	132.96		116.04	0.1395	
40	45.45	118.78		85.92	0.3826	
40	44.43	118.78		85.92	0.3740	
50	18.08	132.96		116.04	0.1360	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	18.54	132.96		116.04	0.1394	
40	45.42	118.78		85.92	0.3824	
40	44.37	118.78		85.92	0.3736	
50	18.08	132.96		116.04	0.1360	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	0.26	132.96		116.04	0.0020	
40	0.51	118.78		85.92	0.0043	
40		118.78	-0.81	85.92		0.0095
50		132.96	-0.50	116.04		0.0043



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Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	18.53	110.80		96.70	0.1673	
40	45.42	98.98		71.60	0.4588	
40	44.45	98.98		71.60	0.4491	
50	18.09	110.80		96.70	0.1633	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30		110.80	-0.71	96.70		0.0074
40		98.98	-0.74	71.60		0.0103
40		98.98	-1.67	71.60		0.0234
50		110.80	-1.38	96.70		0.0142

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
30	18.54	132.96		116.04	0.1395	
40	45.45	118.78		85.92	0.3826	
40	44.43	118.78		85.92	0.3740	
50	18.08	132.96		116.04	0.1360	

Absolute Maximum of the all of the Stress Ratio's 0.4588

Governing Element: Main Shell

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom ch shell	18.3	0.2	...	...	18.5	+ Tens
Bottom ch shell	18.3	0.2	...	...	...	- Comp
Main Shell	44.9	0.3	0.2	...	45.4	+ Tens
Main Shell	44.9	0.3	-0.2	...	...	- Comp
Main Shell	44.9	-0.6	0.2	...	44.5	+ Tens
Main Shell	44.9	-0.6	-0.2	...	...	- Comp
Top ch shell	18.3	-0.3	0.2	...	18.1	+ Tens
Top ch shell	18.3	-0.3	-0.2	...	...	- Comp



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Center of Gravity Calculation: Step: 25 9:37am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	1752.537 mm
Center of Gravity of the Insulation	1752.691 mm
Center of Gravity of the Nozzles	1935.022 mm
Center of Gravity of the Lugs	1903.175 mm
Center of Gravity of the Tubesheet(s)	1752.175 mm
Center of Gravity of the Tubes	1752.175 mm
Center of Gravity of the Baffles	1959.662 mm
Center of Gravity of Bare Shell New and Cold	1752.907 mm
Center of Gravity of Bare Shell Corroded	1753.004 mm
Vessel CG in the Operating Condition	1782.249 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	1791.378 mm
Vessel CG in the Test Condition	1786.423 mm

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Sup. Lug Calcs: Ope:

Lugs: 3 9:37am Sep 9,2026

Support Lug Calculations: Operating Case

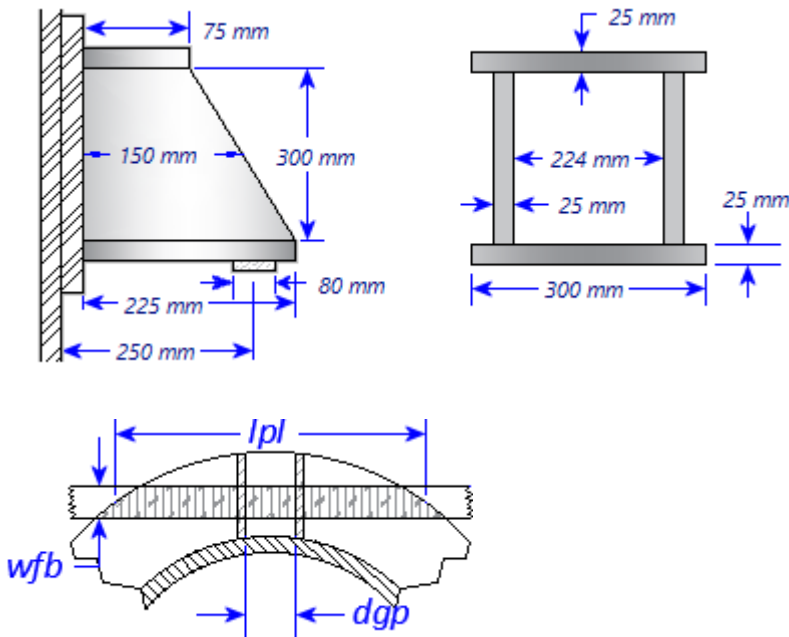
Input Data: Support Lugs

Type of Geometry : Gussets with Top Plate

Number of Support Lugs	Nlug	2
Distance from Vessel OD to Lug Contact Point	Dlug	250.0000 mm
Lug Support Force Bearing Width	Wfb	80.0000 mm
Lug Material	SA-387 11	
Lug Yield Stress		223.00 N./mm ²
Radial Width of bottom Support Lug Plate	Wpl	225.0000 mm
Effective Force Bearing Length	Lpl	300.0000 mm
Thickness of bottom Support Lug Plate	Tpl	25.0000 mm
Distance between Gussets	Dgp	224.0000 mm
Mean Width of Gusset Plate	Wgp	150.0000 mm
Height of Gusset Plate	Hgp	300.0000 mm
Thickness of Gusset Plate	Tgp	25.0000 mm
Radial Width of Top Bar Plate	Wtp	75.0000 mm
Thickness of Top Plate	Ttp	25.0000 mm

Pad Width along Circumference	C11P	160.000 mm
Pad Length along Vessel Axis	C22P	350.000 mm
Pad Thickness	Tpad	25.000 mm

Bolt Material	SA-193 B16	
Bolt Allowable Stress at Design Temperature		170.41 N./mm ²
User Specified Root Area of a single Bolt		1.5500 cm ²
Bolt Diameter		30.00 mm



Results for Support Lugs:

Description: Support Lugs

Overturning Moment at Support Lug	39.0 Kg-m.
Weight Load at the top of one Lug	653.4 Kgf

Force on one Lug, Operating Condition [Flug]:

$$= (W/Nlug + Mlug/(Rlug \cdot Nlug/2))$$



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Sup. Lug Calcs: Ope:

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$$= (1307/2 + 39/(385 \cdot 2/2))$$

$$= 754.62 \text{ Kgf}$$

Top Bar Plate Stress per Bednar p.154 [Stpl]:

$$= 0.75(\text{Flug} \cdot \text{Dlug} \cdot \text{Lpl}) / (\text{Ttp} \cdot \text{Wtp}^2 \cdot \text{Hgp})$$

$$= 0.75(755 \cdot 250 \cdot 300) / (25 \cdot 75^2 \cdot 300)$$

$$= 9.87 \text{ N./mm}^2$$

Required Thickness of Top Plate

1.6593 mm

Bearing Area [Ba]:

$$= \text{Lpl} \cdot \text{Wfb}$$

$$= 300 \cdot 80$$

$$= 240.00 \text{ cm}^2$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 [Spl2]:

$$= \beta_1 \cdot \text{Flug} / \text{Ba} \cdot \text{Wfb}^2 / \text{Tpl}^2 \text{ per Roark \& Young 5th Ed.}$$

$$= 2.11 \cdot 755 / 240 \cdot 80^2 / 25^2$$

$$= 6.65 \text{ N./mm}^2$$

Bottom Plate Required Thickness (Uniform Load)

5.2860 mm

Bottom Plate Required Thickness based on ADM S 3/4 [trAD]:

$$= 0.71 \cdot \text{Dgp} \cdot \sqrt{ (\text{Flug} / (\text{Lpl} \cdot \text{Wfb}) / \text{Spa}) }$$

$$= 0.71 \cdot 224 \cdot \sqrt{ (755 / (300 \cdot 80) / 149) }$$

$$= 7.243 \text{ mm}$$

If using the AD Code recommendations, the force bearing width (Wfb) must be greater than or equal to 1/3 of the bottom plate radial width (Wpl) plus the pad thickness (Padthk), if there is a pad.

Bottom Support Plate Allowable Stress [Spa]:

$$= 2/3 \cdot \text{Ylug}$$

$$= 2/3 \cdot 223$$

$$= 148.67 \text{ N./mm}^2$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (\text{Flug} / 2) / (\text{Wgp} \cdot \text{Tgp})$$

$$= (755 / 2) / (150 \cdot 25)$$

$$= 0.99 \text{ N./mm}^2$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 3.4211 mm

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (\text{Klr}^2 / (2 \cdot \text{Cc}^2))) \text{Fy} / (5/3 + 3 \cdot (\text{Klr} / (8 \cdot \text{Cc}) - (\text{Klr}^3 / (8 \cdot \text{Cc}^3)))$$

$$= (1 - (48^2 / (2 \cdot 125^2))) 223 / (5/3 + 3 \cdot (48 / (8 \cdot 125) - (48^3 / (8 \cdot 125^3)))$$

$$= 114.50 \text{ N./mm}^2$$

Dimension [Gtp]:

$$= 2 \cdot \text{Wgp} - \text{Wpl}$$

$$= 2 \cdot 150 - 225$$

$$= 75.0 \text{ mm}$$

Angle [αG]:

$$= \pi / 2 - \text{atan}((\text{Wpl} - \text{Gtp}) / \text{Hgp})$$

$$= \pi / 2 - \text{atan}((225 - 75) / 300)$$

$$= 63.4 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= \text{Flug} (3 \cdot \text{Dlug} - \text{Wpl}) / (\text{Tgp} \cdot \text{Wpl}^2 \cdot (\text{sin}(\alpha \text{G}))^2)$$

$$= 755 (3 \cdot 250 - 225) / (25 \cdot 225^2 \cdot (\text{sin}(63.4))^2)$$

$$= 3.84 \text{ N./mm}^2$$



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Sup. Lug Calcs: Ope:

Lugs: 3 9:37am Sep 9,2026

Gusset Plate Allowable Compressive Stress [SgaB]:
 $= 18000 / (1 + (1/18000) \cdot (H_{gp} / \sin(\alpha_G) / (0.289 \cdot T_{gp}))^2)$
 $= 18000 / (1 + (1/18000) \cdot (300 / \sin(63.4) / (0.289 \cdot 25))^2)$
 $= 110.84 \text{ N./mm}^2$

There was no uplift. Please choose an appropriate bolt size for this support design.

Input Echo, WRC107/537 Item 1, Description: Support Lugs

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	247.650	mm
Vessel Thickness	Tv	11.113	mm
Design Temperature	T1	430.0	C
Vessel Material		SA-335 P11	
Vessel UNS Number		K11597	
Vessel Cold S.I. Allowable	Smc	118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	98.98	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Rectangular	
Parameter C11	C11	224.00	mm
Parameter C22	C22	300.00	mm
Thickness of Reinforcing Pad	Tpad	25.000	mm
Pad Parameter C11P	C11p	160.000	mm
Pad Parameter C22P	C22p	350.000	mm
Design Internal Pressure	Dp	5900.000	KPa.
Include Pressure Thrust		No	
Vessel Centerline Direction Cosine	Vx	0.000	
Vessel Centerline Direction Cosine	Vy	1.000	
Vessel Centerline Direction Cosine	Vz	0.000	
Nozzle Centerline Direction Cosine	Nx	1.000	
Nozzle Centerline Direction Cosine	Ny	0.000	
Nozzle Centerline Direction Cosine	Nz	0.000	
Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	653.4	Kgf
Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	163.4	Kg-m.
Internal Pressure (SUS)	P	5900.00	KPa.
Include Pressure Thrust		No	
Global Force (OCC)	Fx	37.3	Kgf
Global Force (OCC)	Fy	101.2	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	9.3	Kg-m.
Global Moment (OCC)	Mz	25.3	Kg-m.
Occasional Internal Pressure (OCC)	Pvar	0.00	KPa.

Use Interactive Control

No



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Sup. Lug Calcs: Ope:

Lugs: 3

9:37am

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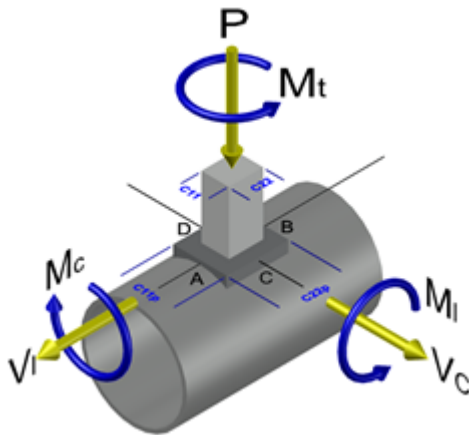
WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No

Compute Pressure Stress per WRC-368 No

Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	653.4	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-163.4	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 5.00 (4.33)

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	1.051	4C !	0.749	(A,B)
N(PHI) / (P/Rm)	1.051	3C !	0.603	(C,D)
M(PHI) / (P)	0.867	2C1 !	0.040	(A,B)
M(PHI) / (P)	0.867	1C !	0.058	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.861	3A !	0.245	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.954	1A !	0.086	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.949	3B !	0.588	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.951	1B !	0.028	(A,B,C,D)
N(x) / (P/Rm)	0.965	3C !	0.603	(A,B)
N(x) / (P/Rm)	0.965	4C !	0.749	(C,D)
M(x) / (P)	0.980	1C1 !	0.059	(A,B)
M(x) / (P)	0.980	2C !	0.039	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.861	4A !	0.487	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	1.116	2A !	0.051	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.949	4B !	0.216	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	1.030	2B !	0.053	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.



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Sup. Lug Calcs: Ope:

Lugs: 3 9:37am

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Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.3	1.3	-1.3	-1.3	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.8	-1.8	-1.8	1.8	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		3.1	-0.5	-3.1	0.5	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.6	0.6	-0.6	-0.6	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		3.2	-3.2	-3.2	3.2	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		3.7	-2.6	-3.7	2.6	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Str. Int.		3.7	2.6	3.7	2.6	0.6	0.6	0.6	0.6

WARNING: Ratio of Pad Radius/Rm (.611) is not between 0.01 and 0.571.

Dimensionless Parameters used : $\Gamma = 16.13$

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	1.218	4C !	1.664	(A,B)
N(PHI) / (P/Rm)	1.218	3C !	0.895	(C,D)
M(PHI) / (P)	0.826	2C1 !	0.014	(A,B)
M(PHI) / (P)	0.826	1C !	0.056	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.793	3A !	0.577	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.894	1A !	0.066	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	1.030	3B !	1.128	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.964	1B !	0.012	(A,B,C,D)
N(x) / (P/Rm)	1.035	3C !	0.895	(A,B)
N(x) / (P/Rm)	1.035	4C !	1.664	(C,D)
M(x) / (P)	1.068	1C1 !	0.028	(A,B)
M(x) / (P)	1.068	2C !	0.030	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.793	4A !	1.714	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	1.142	2A !	0.031	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	1.030	4B !	0.582	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	1.164	2B !	0.022	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.



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Sup. Lug Calcs: Ope:

Lugs: 3 9:37am

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Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		10.8	10.8	-10.8	-10.8	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		13.4	-13.4	-13.4	13.4	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		24.2	-2.7	-24.2	2.7	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		8.4	8.4	-8.4	-8.4	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		20.9	-20.9	-20.9	20.9	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		29.3	-12.5	-29.3	12.5	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Str. Int.		29.3	12.5	29.3	12.5	2.3	2.3	2.3	2.3

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	37.3	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	101.2	Kgf
Circumferential Moment	MC	-9.3	Kg-m.
Longitudinal Moment	ML	-25.3	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : $\Gamma = 5.00$ (4.33)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.3	-0.3	-0.3	0.3
Circ. Memb. ML		0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		0.3	-0.3	-0.3	0.3	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		0.3	-0.1	-0.6	0.1	0.2	-0.2	-0.5	0.4
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.1	-0.1	-0.1	-0.1
Long. Bend. P		-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.1	0.1	-0.1	-0.1
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.2	-0.2	-0.2	0.2



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Sup. Lug Calcs: Ope:

Lugs: 3 9:37am

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Long. Memb. ML	0.1	0.1	-0.1	-0.1	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.5	-0.5	-0.5	0.5	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	0.4	-0.3	-0.7	0.5	0.1	-0.1	-0.4	0.1
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Str. Int.	0.4	0.3	0.7	0.5	0.2	0.2	0.5	0.4

WARNING: Ratio of Pad Radius/Rm (.611) is not between 0.01 and 0.571.

Dimensionless Parameters used : Gamma = 16.13

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.6	-0.6	-0.6	-0.6	-0.3	-0.3	-0.3	-0.3
Circ. Bend. P		-0.5	0.5	-0.5	0.5	-1.9	1.9	-1.9	1.9
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.3	0.3	-0.3	-0.3
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	4.7	-4.7	-4.7	4.7
Circ. Memb. ML		1.7	1.7	-1.7	-1.7	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		2.1	-2.1	-2.1	2.1	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.7	-0.5	-4.8	0.3	2.8	-2.8	-7.1	5.9
Long. Memb. P		-0.3	-0.3	-0.3	-0.3	-0.6	-0.6	-0.6	-0.6
Long. Bend. P		-0.9	0.9	-0.9	0.9	-1.0	1.0	-1.0	1.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	1.0	1.0	-1.0	-1.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	1.7	-1.7	-1.7	1.7
Long. Memb. ML		1.3	1.3	-1.3	-1.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		3.2	-3.2	-3.2	3.2	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		3.3	-1.3	-5.8	2.6	1.2	-0.3	-4.3	1.2
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Str. Int.		3.3	1.3	5.8	2.6	2.8	2.9	7.1	5.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		20.0	25.9	20.0	25.9	20.0	25.9	20.0	25.9
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		20.0	25.9	20.0	25.9	20.0	25.9	20.0	25.9
Circ. Pl (SUS)		1.3	1.3	-1.3	-1.3	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		0.1	0.1	-0.3	-0.3	-0.0	-0.0	-0.1	-0.1



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Circ. Pl(TOTAL)	1.4	1.4	-1.5	-1.5	-0.0	-0.0	-0.1	-0.1
Circ. Q (SUS)	1.8	-1.8	-1.8	1.8	0.0	0.0	-0.0	0.0
Circ. Q (OCC)	0.2	-0.2	-0.4	0.4	0.2	-0.2	-0.4	0.4
Circ. Q (TOTAL)	2.0	-2.0	-2.2	2.2	0.2	-0.2	-0.4	0.4
Long. Pm (SUS)	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
Long. Pl (SUS)	0.6	0.6	-0.6	-0.6	0.0	0.0	-0.0	-0.0
Long. Pl (OCC)	0.0	0.0	-0.1	-0.1	0.0	0.0	-0.1	-0.1
Long. Pl(TOTAL)	0.6	0.6	-0.7	-0.7	0.0	0.0	-0.1	-0.1
Long. Q (SUS)	3.2	-3.2	-3.2	3.2	0.0	0.0	-0.0	0.0
Long. Q (OCC)	0.4	-0.4	-0.6	0.6	0.1	-0.1	-0.2	0.2
Long. Q (TOTAL)	3.5	-3.5	-3.8	3.8	0.1	-0.1	-0.2	0.2
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear Pl (OCC)	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear Pl(TOTAL)	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	20.0	25.9	20.0	25.9	20.0	25.9	20.0	25.9
Pm (SUS+OCC)	20.0	25.9	20.0	25.9	20.0	25.9	20.0	25.9
Pm+Pl (SUS)	21.3	27.2	18.7	24.6	20.0	25.9	20.0	25.9
Pm+Pl (SUS+OCC)	21.4	27.3	18.4	24.3	20.0	25.9	19.9	25.8
Pm+Pl+Q (Total)	23.4	25.3	16.3	26.5	20.2	25.7	19.5	26.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.89	98.98	Passed
Pm (SUS+OCC)	25.89	118.78	Passed
Pm+Pl (SUS)	27.18	148.47	Passed
Pm+Pl (SUS+OCC)	27.32	178.17	Passed
Pm+Pl+Q (TOTAL)	26.51	296.95	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		89.4	95.3	89.4	95.3	89.4	95.3	89.4	95.3
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		89.4	95.3	89.4	95.3	89.4	95.3	89.4	95.3
Circ. Pl (SUS)		10.8	10.8	-10.8	-10.8	0.0	0.0	-0.0	-0.0



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Circ. Pl (OCC)	1.1	1.1	-2.2	-2.2	-0.0	-0.0	-0.6	-0.6
Circ. Pl(TOTAL)	11.8	11.8	-13.0	-13.0	-0.0	-0.0	-0.6	-0.6
Circ. Q (SUS)	13.4	-13.4	-13.4	13.4	0.0	0.0	-0.0	0.0
Circ. Q (OCC)	1.6	-1.6	-2.5	2.5	2.8	-2.8	-6.5	6.5
Circ. Q (TOTAL)	15.0	-15.0	-16.0	16.0	2.8	-2.8	-6.5	6.5
Long. Pm (SUS)	44.7	44.7	44.7	44.7	44.7	44.7	44.7	44.7
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	44.7	44.7	44.7	44.7	44.7	44.7	44.7	44.7
Long. Pl (SUS)	8.4	8.4	-8.4	-8.4	0.0	0.0	-0.0	-0.0
Long. Pl (OCC)	1.0	1.0	-1.6	-1.6	0.4	0.4	-1.6	-1.6
Long. Pl(TOTAL)	9.4	9.4	-10.0	-10.0	0.4	0.4	-1.6	-1.6
Long. Q (SUS)	20.9	-20.9	-20.9	20.9	0.0	0.0	-0.0	0.0
Long. Q (OCC)	2.3	-2.3	-4.2	4.2	0.7	-0.7	-2.7	2.7
Long. Q (TOTAL)	23.2	-23.2	-25.1	25.1	0.7	-0.7	-2.7	2.7
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear Pl (OCC)	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear Pl(TOTAL)	0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	89.4	95.3	89.4	95.3	89.4	95.3	89.4	95.3
Pm (SUS+OCC)	89.4	95.3	89.4	95.3	89.4	95.3	89.4	95.3
Pm+Pl (SUS)	100.1	106.0	78.6	84.5	89.4	95.3	89.4	95.3
Pm+Pl (SUS+OCC)	101.2	107.1	76.4	82.3	89.4	95.3	88.8	94.7
Pm+Pl+Q (Total)	116.3	92.1	60.4	98.3	92.2	92.5	82.3	101.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	95.28	98.98	Passed
Pm (SUS+OCC)	95.28	118.78	Passed
Pm+Pl (SUS)	106.03	148.47	Passed
Pm+Pl (SUS+OCC)	107.13	178.17	Passed
Pm+Pl+Q (TOTAL)	116.27	296.95	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

[Review notes about nozzle loadings and supports in the Warnings report.](#)

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Support Lug Calculations: Test Case

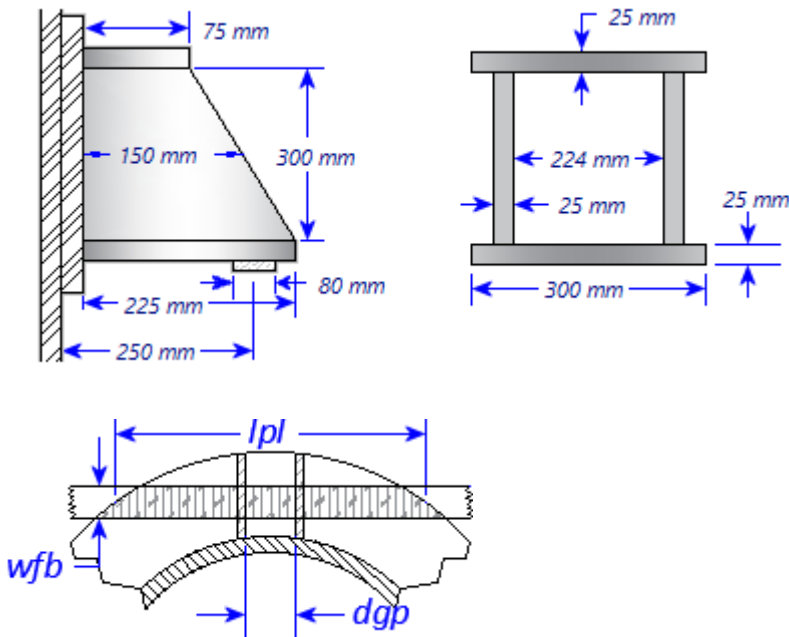
Input Data: Support Lugs

Type of Geometry : Gussets with Top Plate

Number of Support Lugs	Nlug	2
Distance from Vessel OD to Lug Contact Point	Dlug	250.0000 mm
Lug Support Force Bearing Width	Wfb	80.0000 mm
Lug Material	SA-387 11	
Lug Yield Stress		309.99 N./mm ²
Radial Width of bottom Support Lug Plate	Wpl	225.0000 mm
Effective Force Bearing Length	Lpl	300.0000 mm
Thickness of bottom Support Lug Plate	Tpl	25.0000 mm
Distance between Gussets	Dgp	224.0000 mm
Mean Width of Gusset Plate	Wgp	150.0000 mm
Height of Gusset Plate	Hgp	300.0000 mm
Thickness of Gusset Plate	Tgp	25.0000 mm
Radial Width of Top Bar Plate	Wtp	75.0000 mm
Thickness of Top Plate	Ttp	25.0000 mm

Pad Width along Circumference	C11P	160.000 mm
Pad Length along Vessel Axis	C22P	350.000 mm
Pad Thickness	Tpad	25.000 mm

Bolt Material	SA-193 B16	
Bolt Allowable Stress at Design Temperature		172.00 N./mm ²
User Specified Root Area of a single Bolt		1.5500 cm ²
Bolt Diameter		30.00 mm



Results for Support Lugs:

Description: Support Lugs

Overturning Moment at Support Lug	0.0 Kg-m.
Weight Load at the top of one Lug	633.7 Kg _f

Force on one Lug, Test Condition [Flug]:

$$= (W/Nlug + Mlug/(Rlug \cdot Nlug/2))$$



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$$= (1267/2 + 0.043/(385 \cdot 2/2))$$

$$= 633.82 \text{ Kgf}$$

Top Bar Plate Stress per Bednar p.154 [Stpl]:

$$= 0.75(\text{Flug} \cdot \text{Dlug} \cdot \text{Lpl}) / (\text{Ttp} \cdot \text{Wtp}^2 \cdot \text{Hgp})$$

$$= 0.75(634 \cdot 250 \cdot 300) / (25 \cdot 75^2 \cdot 300)$$

$$= 8.29 \text{ N./mm}^2$$

Required Thickness of Top Plate

1.5875 mm

Bearing Area [Ba]:

$$= \text{Lpl} \cdot \text{Wfb}$$

$$= 300 \cdot 80$$

$$= 240.00 \text{ cm}^2$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 [Spl2]:

$$= \beta_1 \cdot \text{Flug}/\text{Ba} \cdot \text{Wfb}^2 / \text{Tpl}^2 \text{ per Roark \& Young 5th Ed.}$$

$$= 2.11 \cdot 634/240 \cdot 80^2/25^2$$

$$= 5.58 \text{ N./mm}^2$$

Bottom Plate Required Thickness (Uniform Load)

4.1089 mm

Bottom Plate Required Thickness based on ADM S 3/4 [trAD]:

$$= 0.71 \cdot \text{Dgp} \cdot \sqrt{ (\text{Flug} / (\text{Lpl} \cdot \text{Wfb}) / \text{Spa}) }$$

$$= 0.71 \cdot 224 \cdot \sqrt{ (634 / (300 \cdot 80) / 207) }$$

$$= 5.630 \text{ mm}$$

If using the AD Code recommendations, the force bearing width (Wfb) must be greater than or equal to 1/3 of the bottom plate radial width (Wpl) plus the pad thickness (Padthk), if there is a pad.

Bottom Support Plate Allowable Stress [Spa]:

$$= 2/3 \cdot \text{Ylug}$$

$$= 2/3 \cdot 310$$

$$= 206.66 \text{ N./mm}^2$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (\text{Flug}/2) / (\text{Wgp} \cdot \text{Tgp})$$

$$= (634/2) / (150 \cdot 25)$$

$$= 0.83 \text{ N./mm}^2$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 3.0760 mm

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (\text{Klr}^2 / (2 \cdot \text{Cc}^2))) \text{Fy} / (5/3 + 3 \cdot (\text{Klr}) / (8 \cdot \text{Cc}) - (\text{Klr}^3) / (8 \cdot \text{Cc}^3))$$

$$= (1 - (48)^2 / (2 \cdot 114^2)) 310 /$$

$$(5/3 + 3 \cdot (48) / (8 \cdot 114) - (48^3) / (8 \cdot 114^3))$$

$$= 155.64 \text{ N./mm}^2$$

Dimension [Gtp]:

$$= 2 \cdot \text{Wgp} - \text{Wpl}$$

$$= 2 \cdot 150 - 225$$

$$= 75.0 \text{ mm}$$

Angle [αG]:

$$= \pi / 2 - \text{atan}((\text{Wpl} - \text{Gtp}) / \text{Hgp})$$

$$= \pi / 2 - \text{atan}((225 - 75) / 300)$$

$$= 63.4 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= \text{Flug}(3 \cdot \text{Dlug} - \text{Wpl}) / (\text{Tgp} \cdot \text{Wpl}^2 \cdot (\text{sind}(\alpha G))^2)$$

$$= 634 (3 \cdot 250 - 225) / (25 \cdot 225^2 \cdot (\text{sind}(63.4))^2)$$

$$= 3.22 \text{ N./mm}^2$$



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Gusset Plate Allowable Compressive Stress [SgaB]:
 $= 18000 / (1 + (1/18000) \cdot (H_{gp} / \sin(\alpha_G) / (0.289 \cdot T_{gp}))^2)$
 $= 18000 / (1 + (1/18000) \cdot (300 / \sin(63.4) / (0.289 \cdot 25))^2)$
 $= 110.84 \text{ N./mm}^2$

There was no uplift. Please choose an appropriate bolt size for this support design.

Input Echo, WRC107/537 Item 1, Description: Support Lugs

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	247.650	mm
Vessel Thickness	Tv	11.113	mm
Design Temperature	T1	21.1	C
Vessel Material		SA-335 P11	
Vessel UNS Number		K11597	
Vessel Cold S.I. Allowable	Smc	184.50	N./mm ²
Vessel Hot S.I. Allowable	Smh	184.50	N./mm ²
Attachment Type	Type	Rectangular	
Parameter C11	C11	224.00	mm
Parameter C22	C22	300.00	mm
Thickness of Reinforcing Pad	Tpad	25.000	mm
Pad Parameter C11P	C11p	160.000	mm
Pad Parameter C22P	C22p	350.000	mm
Design Internal Pressure	Dp	5900.000	KPa.
Include Pressure Thrust		No	
Vessel Centerline Direction Cosine	Vx	0.000	
Vessel Centerline Direction Cosine	Vy	1.000	
Vessel Centerline Direction Cosine	Vz	0.000	
Nozzle Centerline Direction Cosine	Nx	1.000	
Nozzle Centerline Direction Cosine	Ny	0.000	
Nozzle Centerline Direction Cosine	Nz	0.000	
Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	443.7	Kgf
Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	110.9	Kg-m.
Internal Pressure (SUS)	P	5900.00	KPa.
Include Pressure Thrust		No	
Global Force (OCC)	Fx	0.0	Kgf
Global Force (OCC)	Fy	190.1	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	47.5	Kg-m.
Occasional Internal Pressure (OCC)	Pvar	1782.49	KPa.
Use Interactive Control		No	
WRC107 Version	Version	March	1979
Include Pressure Stress Indices per Div. 2		No	



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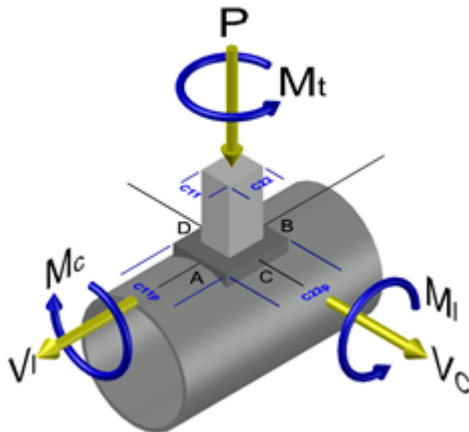
Compute Pressure Stress per WRC-368

No

Local Loads applied at end of Nozzle/Attachment

No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	443.7	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-110.9	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 5.00 (4.33)

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	1.051	4C !	0.749	(A,B)
N(PHI) / (P/Rm)	1.051	3C !	0.603	(C,D)
M(PHI) / (P)	0.867	2C1 !	0.040	(A,B)
M(PHI) / (P)	0.867	1C !	0.058	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.861	3A !	0.245	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.954	1A !	0.086	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.949	3B !	0.588	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.951	1B !	0.028	(A,B,C,D)
N(x) / (P/Rm)	0.965	3C !	0.603	(A,B)
N(x) / (P/Rm)	0.965	4C !	0.749	(C,D)
M(x) / (P)	0.980	1C1 !	0.059	(A,B)
M(x) / (P)	0.980	2C !	0.039	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.861	4A !	0.487	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	1.116	2A !	0.051	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.949	4B !	0.216	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	1.030	2B !	0.053	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



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Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.9	0.9	-0.9	-0.9	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.2	-1.2	-1.2	1.2	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.1	-0.4	-2.1	0.4	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.4	0.4	-0.4	-0.4	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		2.1	-2.1	-2.1	2.1	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.5	-1.8	-2.5	1.8	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Str. Int.		2.5	1.8	2.5	1.8	0.4	0.4	0.4	0.4

WARNING: Ratio of Pad Radius/Rm (.611) is not between 0.01 and 0.571.

Dimensionless Parameters used : Gamma = 16.13

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	1.218	4C !	1.664	(A,B)
N(PHI) / (P/Rm)	1.218	3C !	0.895	(C,D)
M(PHI) / (P)	0.826	2C1 !	0.014	(A,B)
M(PHI) / (P)	0.826	1C !	0.056	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.793	3A !	0.577	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.894	1A !	0.066	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	1.030	3B !	1.128	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.964	1B !	0.012	(A,B,C,D)
N(x) / (P/Rm)	1.035	3C !	0.895	(A,B)
N(x) / (P/Rm)	1.035	4C !	1.664	(C,D)
M(x) / (P)	1.068	1C1 !	0.028	(A,B)
M(x) / (P)	1.068	2C !	0.030	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.793	4A !	1.714	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	1.142	2A !	0.031	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	1.030	4B !	0.582	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	1.164	2B !	0.022	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



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Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		7.3	7.3	-7.3	-7.3	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		9.1	-9.1	-9.1	9.1	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		16.4	-1.8	-16.4	1.8	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		5.7	5.7	-5.7	-5.7	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		14.2	-14.2	-14.2	14.2	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		19.9	-8.5	-19.9	8.5	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Str. Int.		19.9	8.5	19.9	8.5	1.5	1.5	1.5	1.5

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	190.1	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-47.5	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 5.00 (4.33)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.4	0.4	-0.4	-0.4	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		0.5	-0.5	-0.5	0.5	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		0.9	-0.2	-0.9	0.2	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		0.9	-0.9	-0.9	0.9	-0.0	-0.0	-0.0	-0.0



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Tot. Long. Str.	1.1	-0.8	-1.1	0.8	0.0	0.0	0.0	0.0
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Str. Int.	1.1	0.8	1.1	0.8	0.2	0.2	0.2	0.2

WARNING: Ratio of Pad Radius/Rm (.611) is not between 0.01 and 0.571.

Dimensionless Parameters used : Gamma = 16.13

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		3.1	3.1	-3.1	-3.1	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		3.9	-3.9	-3.9	3.9	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		7.0	-0.8	-7.0	0.8	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		2.4	2.4	-2.4	-2.4	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		6.1	-6.1	-6.1	6.1	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		8.5	-3.6	-8.5	3.6	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Str. Int.		8.5	3.6	8.5	3.6	0.7	0.7	0.7	0.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		20.0	25.9	20.0	25.9	20.0	25.9	20.0	25.9
Circ. Pm (OCC)		6.0	7.8	6.0	7.8	6.0	7.8	6.0	7.8
Circ. Pm(TOTAL)		26.0	33.7	26.0	33.7	26.0	33.7	26.0	33.7
Circ. Pl (SUS)		0.9	0.9	-0.9	-0.9	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		0.4	0.4	-0.4	-0.4	0.0	0.0	-0.0	-0.0
Circ. Pl(TOTAL)		1.2	1.2	-1.2	-1.2	0.0	0.0	0.0	0.0
Circ. Q (SUS)		1.2	-1.2	-1.2	1.2	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		0.5	-0.5	-0.5	0.5	0.0	0.0	-0.0	0.0



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Circ. Q (TOTAL)	1.8	-1.8	-1.8	1.8	0.0	0.0	0.0	0.0
Long. Pm (SUS)	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
Long. Pm (OCC)	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Long. Pm(TOTAL)	13.0	13.0	13.0	13.0	13.0	13.0	13.0	13.0
Long. P1 (SUS)	0.4	0.4	-0.4	-0.4	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	0.2	0.2	-0.2	-0.2	0.0	0.0	-0.0	-0.0
Long. P1(TOTAL)	0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Long. Q (SUS)	2.1	-2.1	-2.1	2.1	0.0	0.0	-0.0	0.0
Long. Q (OCC)	0.9	-0.9	-0.9	0.9	0.0	0.0	-0.0	0.0
Long. Q (TOTAL)	3.1	-3.1	-3.1	3.1	0.0	0.0	0.0	0.0
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	20.0	25.9	20.0	25.9	20.0	25.9	20.0	25.9
Pm (SUS+OCC)	26.0	33.7	26.0	33.7	26.0	33.7	26.0	33.7
Pm+P1 (SUS)	20.9	26.8	19.1	25.0	20.0	25.9	20.0	25.9
Pm+P1 (SUS+OCC)	27.3	35.0	24.8	32.5	26.0	33.7	26.0	33.7
Pm+P1+Q (Total)	29.0	33.2	23.0	34.2	26.0	33.7	26.0	33.7

Vessel Stress Summation Comparison (N/mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.89	184.50	Passed
Pm (SUS+OCC)	33.71	221.40	Passed
Pm+P1 (SUS)	26.76	276.74	Passed
Pm+P1 (SUS+OCC)	34.96	332.09	Passed
Pm+P1+Q (TOTAL)	34.22	553.49	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N/mm²)

Type of Stress	Load	Stress Intensity Values at						
		Au	Al	Bu	Bl	Cu	Cl	Du
Circ. Pm (SUS)		89.4	95.3	89.4	95.3	89.4	95.3	89.4
Circ. Pm (OCC)		27.0	28.8	27.0	28.8	27.0	28.8	27.0
Circ. Pm(TOTAL)		116.4	124.1	116.4	124.1	116.4	124.1	116.4
Circ. P1 (SUS)		7.3	7.3	-7.3	-7.3	0.0	0.0	-0.0
Circ. P1 (OCC)		3.1	3.1	-3.1	-3.1	0.0	0.0	-0.0
Circ. P1(TOTAL)		10.4	10.4	-10.4	-10.4	0.0	0.0	0.0
Circ. Q (SUS)		9.1	-9.1	-9.1	9.1	0.0	0.0	-0.0



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Circ. Q (OCC)	3.9	-3.9	-3.9	3.9	0.0	0.0	-0.0	0.0
Circ. Q (TOTAL)	13.0	-13.0	-13.0	13.0	0.0	0.0	0.0	0.0
Long. Pm (SUS)	44.7	44.7	44.7	44.7	44.7	44.7	44.7	44.7
Long. Pm (OCC)	13.5	13.5	13.5	13.5	13.5	13.5	13.5	13.5
Long. Pm(TOTAL)	58.2	58.2	58.2	58.2	58.2	58.2	58.2	58.2
Long. P1 (SUS)	5.7	5.7	-5.7	-5.7	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	2.4	2.4	-2.4	-2.4	0.0	0.0	-0.0	-0.0
Long. P1(TOTAL)	8.1	8.1	-8.1	-8.1	0.0	0.0	0.0	0.0
Long. Q (SUS)	14.2	-14.2	-14.2	14.2	0.0	0.0	-0.0	0.0
Long. Q (OCC)	6.1	-6.1	-6.1	6.1	0.0	0.0	-0.0	0.0
Long. Q (TOTAL)	20.3	-20.3	-20.3	20.3	0.0	0.0	0.0	0.0
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	89.4	95.3	89.4	95.3	89.4	95.3	89.4	95.3
Pm (SUS+OCC)	116.4	124.1	116.4	124.1	116.4	124.1	116.4	124.1
Pm+P1 (SUS)	96.7	102.6	82.1	88.0	89.4	95.3	89.4	95.3
Pm+P1 (SUS+OCC)	126.8	134.5	106.0	113.6	116.4	124.1	116.4	124.1
Pm+P1+Q (Total)	139.8	121.5	92.9	126.7	116.4	124.1	116.4	124.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	95.28	184.50	Passed
Pm (SUS+OCC)	124.07	221.40	Passed
Pm+P1 (SUS)	102.58	276.74	Passed
Pm+P1 (SUS+OCC)	134.50	332.09	Passed
Pm+P1+Q (TOTAL)	139.84	553.49	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

[Review notes about nozzle loadings and supports in the Warnings report.](#)

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Nozzle Calcs.: N1-Tube Inlet

Nozl: 9

9:37am

Sep 9, 2026

Input, Nozzle Desc: N1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	2431.669	KPa.
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	270	°C
Parent Material		SA-335 P11	
Parent Allowable Stress at Temperature	Sv	110.80	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	247.65	mm
Design Length of Section	L	299.9689	mm
Shell Finished (Minimum) Thickness	t	11.1125	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		277.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	110.80	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	11.0744	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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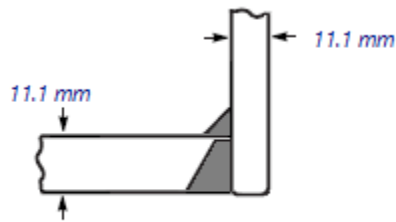
FileName : 160-C-Vertical

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Nozzle Calcs.: N1-Tube Inlet

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Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.503 in.
Actual Thickness Used in Calculation	0.436 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2432 \cdot 127 / (111 \cdot 1 - 0.6 \cdot 2432)$
 $= 2.8207 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2432 \cdot 22.1 / (111 \cdot 1 - 0.6 \cdot 2432)$
 $= 0.4913 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	88.3524 mm
Parallel to Vessel Wall, opening length	d	44.1762 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.1860 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N1-Tube Inlet.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4913 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 5.8207 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 5.8207 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(5.82, 5.5)]$
 $= 5.8207 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.49, 5.82)$
 $= 5.8207 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.074 = 9.690 \text{ mm} \rightarrow \text{OK}$



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Nozzle Calcs.: N1-Tube Inlet

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Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	45.5,	Allowable	:	110.8 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	240.5 N./mm ²	Passed
Occasional	:	2.8,	Allowable	:	147.4 N./mm ²	Passed
Shear	:	31.4,	Allowable	:	77.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 9.69$, $t_r = 0.46$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.069$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2431.67 / 10200.00 = 0.238$

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds T_{min} 8.0744 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6521 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.7298 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 164.8423 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.0000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	247.650 mm
Shell Thickness	11.1125 mm
Shell Stress Concentration Factor	1.000
Vessel Material	SA-335 P11
Vessel Cold S.I. Allowable	Smc 118.00 N./mm ²
Vessel Hot S.I. Allowable	Smh 110.80 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID
Nozzle Corrosion Allowance	3.0000 mm
Nozzle Diameter	38.176 mm
Nozzle Thickness	11.0744 mm
Nozzle Stress Concentration Factor	1.000



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Nozzle Calcs.: N1-Tube Inlet

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Nozzle Material	SA-335 P11
Nozzle Cold S.I. Allowable	SNmc 118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh 110.80 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	2431.67	KPa.
Radial Load	P	-458.90	Kgf
Circumferential Shear	Vc	331.43	Kgf
Longitudinal Shear	Vl	331.43	Kgf
Circumferential Moment	Mc	50.97	Kg-m.
Longitudinal Moment	Ml	50.97	Kg-m.
Torsional Moment	Mt	58.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.447) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{130.881(11.113 - 3.0)}$$

$$= 167.855 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	261.763 mm
Nozzle Outside Diameter	(d) =	60.325 mm
Vessel Thickness used	(T) =	8.113 mm
Nozzle Thickness used	(t) =	8.074 mm
T / t	=	1.005
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.309$$

Nr/P	=	0.111
Mr/P	=	0.157
M0/P	=	0.053
N0/P	=	0.112
MrD/Mc	=	0.269
NrDL/Mc	=	0.146
M0d/Mc	=	0.102
N0DL/Mc	=	0.047
MrD/Ml	=	0.165
NrDL/Ml	=	0.099
M0D/Ml	=	0.061
N0DL/Ml	=	0.133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	21	-21	21	-21
Outplane Membrane (Mc)	0	0	0	0



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Nozzle Calcs.: N1-Tube Inlet

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Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-46	46	46	-46
Normal Pressure Stress	36	39	36	39

Outplane Stress Summary	3	54	129	-4
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	64	-64	64	-64
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-12	-12	12	12
Inplane Bending (ML)	-124	124	124	-124
Inplane Pressure Stress	18	18	18	18

Inplane Stress Summary	-46	73	227	-150
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	12	12	12	12

Shear Stress Summary	16	16	8	8
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	59	83	228	150

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	21	-21	21	-21
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-76	76	76	-76
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18	18	18	18

Outplane Stress Summary	-34	75	130	-66
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	64	-64	64	-64
Inplane Membrane (Mc)	-18	-18	18	18
Inplane Bending (Mc)	-203	203	203	-203
Inplane Membrane (ML)	0	0	0	0



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Nozzle Calcs.: N1-Tube Inlet

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Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	36	39	36	39

Inplane Stress Summary	-112	167	330	-202
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	12	12	12	12

Shear Stress Summary	8	8	16	16
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	113	168	331	204
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	9	6	9

Hoop Stress Summary	-2	0	31	34
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	42	-42	42	-42
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-32	-32	32	32
Axial Bending (ML)	-87	87	87	-87
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-71	19	169	-90
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12

Shear Stress Summary	16	16	8	8
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: N1-Tube Inlet

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(Stress Intensities	Outside	Inside	Outside	Inside
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Two * Max Shear Stress	76	38	169	125
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	9	6	9

Hoop Stress Summary	8	11	20	23
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	42	-42	42	-42
Axial Membrane (Mc)	-32	-32	32	32
Axial Bending (Mc)	-149	149	149	-149
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-133	81	231	-152
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	12	12	12	12

Shear Stress Summary	8	8	16	16
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	142	82	232	178
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	36	39	36	39	36	39	36	39
Circ. Pl (SUS)	-9	-9	23	23	-11	-11	25	25
Circ. Q (SUS)	-25	25	67	-67	-139	139	267	-267
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	-5	-5	19	19	2	2	12	12
Long. Q (SUS)	-60	60	188	-188	-55	55	97	-97



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Nozzle Calcs.: N1-Tube Inlet

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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	36.0	39.0	36.0	39.0	36.0	39.0	36.0	39.0
Pm+P1 (SUS)	28.1	30.9	59.7	62.6	27.2	29.7	61.5	64.5
Pm+P1+Q (Total)	58.5	82.4	225.6	151.4	114.8	167.7	329.3	204.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.00	110.80	Passed
Pm+P1 (SUS)	64.46	166.20	Passed
Pm+P1+Q (TOTAL)	329.27	332.40	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	9	6	9	6	9	6	9
Circ. P1 (SUS)	-9	-9	23	23	2	2	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. P1 (SUS)	-29	-29	35	35	-29	-29	35	35
Long. Q (SUS)	-45	45	129	-129	-107	107	191	-191
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	6.0	9.0	6.0	9.0	6.0	9.0	6.0	9.0
Pm+P1 (SUS)	26.7	27.2	39.5	40.0	34.9	37.9	38.8	38.9
Pm+P1+Q (Total)	75.2	37.2	167.5	124.0	141.9	81.9	230.2	176.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	110.80	Passed
Pm+P1 (SUS)	40.00	166.20	Passed
Pm+P1+Q (TOTAL)	230.21	332.40	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Nozzle Calcs.: N3-Shell inlet

Nozl: 10

9:37am

Sep 9, 2026

Input, Nozzle Desc: N3-Shell inlet

From: 40

Pressure for Reinforcement Calculations	P	5920.094	KPa.
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	430	°C
Parent Material		SA-335 P11	
Parent Allowable Stress at Temperature	Sv	98.98	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	247.65	mm
Design Length of Section	L	2550.0002	mm
Shell Finished (Minimum) Thickness	t	11.1125	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		977.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	98.98	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	80.48	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



PV Elite 28 Licensee: SL User

FileName : 160-C-Vertical

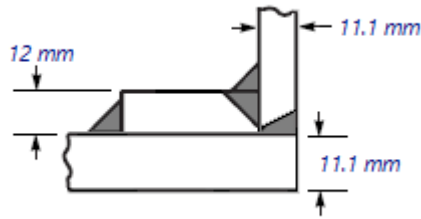
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Nozzle Calcs.: N3-Shell inlet

Noz1: 10

9:37am

Sep 9, 2026



Reinforcement CALCULATION, Description: N3-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.624 in.
Actual Thickness Used in Calculation	0.438 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5920 \cdot 127 / (99 \cdot 1 - 0.6 \cdot 5920)$
 $= 7.8682 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5920 \cdot 49 / (99 \cdot 1 - 0.6 \cdot 5920)$
 $= 3.0415 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5061 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	196.0992 mm
Parallel to Vessel Wall, opening length	d	98.0496 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		20.2812 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 99 / 99)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 80.5 / 99)$
 $= 0.813$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 0.81)$
 $= 0.813$



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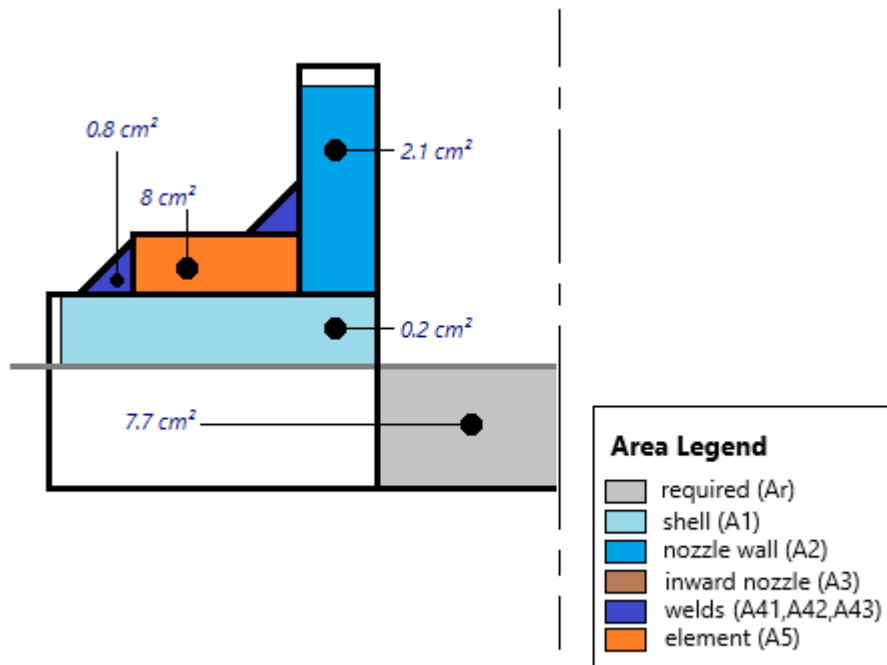


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	7.715	1.198	NA
Area in Shell	A1	0.240	5.559	NA
Area in Nozzle Wall	A2	2.062	3.091	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.789	0.789	NA
Area in Element	A5	7.981	7.981	NA
TOTAL AREA AVAILABLE	Atot	11.072	17.420	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 161.6909 12.0000 mm
Based on given Pad Diameter: 200.0000 6.9523 mm
Based on Shell or Nozzle Thickness: 165.4757 11.1125 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (98 \cdot 7.87 \cdot 1 + 2 \cdot 8.13 \cdot 7.87 \cdot 1 \cdot (1 - 1))$$

$$= 7.715 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 98 (1 \cdot 8.11 - 1 \cdot 7.87) - 2 \cdot 8.13$$

$$(1 \cdot 8.11 - 1 \cdot 7.87) \cdot (1 - 1)$$



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$$= 0.240 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 20.3) \cdot (8.13 - 3.04) \cdot 1$$

$$= 2.062 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_{r \text{ Lost}}) \cdot F_{r3} + ((W_i - c_{an}/0.707)^2 - A_{r \text{ Lost}}) \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= (0.97) \cdot 0.81 + (0) \cdot 1 + 0^2 \cdot 0.81$$

$$= 0.789 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (196 - 114) \cdot 12 \cdot 0.81$$

$$= 7.981 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 6.0415 \text{ mm}$

Wall Thickness per UG16(b), $t_{r16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $t_{rb1} = 10.8682 \text{ mm}$

Wall Thickness $t_{b1} = \max(t_{rb1}, t_{r16b}) = 10.8682 \text{ mm}$

Wall Thickness $t_{b2} = \max(t_{rb2}, t_{r16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $t_{b3} = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[8.26, \max(10.9, 5.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(6.04, 8.26)$$

$$= 8.2578 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 13.2,	Allowable	: 99.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 258.0 N./mm ²	Passed
Occasional	: 16.5,	Allowable	: 131.6 N./mm ²	Passed
Shear	: 7.4,	Allowable	: 69.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9.73$, $t_r = 2.54$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.38$, Temp. Reduction = $58 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $-29 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-48 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.73$, $t_r = 2.54$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.38$, Temp. Reduction = $58 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $-29 \text{ }^\circ\text{C}$

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit $-46 \text{ }^\circ\text{C}$



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Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 11.1, tr = 6.56, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.81$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -42 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 9.73, tr = 2.54, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.38$, Temp. Reduction = 58 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C

Gov. MDMT of the Reinforcement Pad : -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $5920.09/15310.00 = 0.387$

Weld Size Calculations, Description: N3-Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 8.1252 mm

Intermediate Calc. for pad/shell Welds TminPad 9.7000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6876 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$4.8500 = 0.5 \cdot T_{minPad}$	$5.6560 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= (Ar - A1) \cdot Sv$

$= (7.71 - 0.24) \cdot 99$

$= 7544.84$ Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$

$= (2.06 + 7.98 + 0.79 - 0 \cdot 1) \cdot 99$

$= 10933.42$ Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$

$= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 80.5$

$= 7220$ Kgf

Shear, Pad Element Weld [Spew]:

$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$



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$$= (3.14/2.0) \cdot 200 \cdot 8 \cdot 0.49 \cdot 80.5$$

$$= 10107. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot \text{Seg}$$

$$= (3.14/2) \cdot 114 \cdot 12 \cdot 0.74 \cdot 99$$

$$= 16092. \text{ Kg}$$

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNGW}) = (10107 + 0) = 10107 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 10106 Kg, must exceed W = 7544 Kg or W1 = 10933 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6072 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 13.9773 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9398 mm

Input Echo, WRC297 Item 1, Description: N3-Shell inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	247.650	mm
Shell Thickness	11.1125	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-335 P11	
Vessel Cold S.I. Allowable	Smc 118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 98.98	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	92.050	mm
Nozzle Thickness	11.1252	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-335 P11	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 98.98	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	5920.09	KPa.
Radial Load	P	-917.81	Kgf
Circumferential Shear	Vc	662.86	Kgf
Longitudinal Shear	Vl	662.86	Kgf



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Circumferential Moment	Mc	0.20	Kg-m.
Longitudinal Moment	Ml	0.20	Kg-m.
Torsional Moment	Mt	0.23	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{136.881(11.113 - 3.0)}$$

$$= 224.267 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	273.763 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	20.112 mm
Nozzle Thickness used	(t) =	8.125 mm
T / t	=	2.475
d / t	=	14.067

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.540$$

Nr/P	=	0.119
Mr/P	=	0.107
M0/P	=	0.086
N0/P	=	0.173
MrD/Mc	=	0.191
NrDL/Mc	=	0.165
M0d/Mc	=	0.169
N0DL/Mc	=	0.088
MrD/Ml	=	0.146
NrDL/Ml	=	0.128
M0D/Ml	=	0.133
N0DL/Ml	=	0.205

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	34	40	34	40
Outplane Stress Summary	49	32	49	32

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (Ml)	0	0	0	0



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Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	17	17	17	17
Inplane Stress Summary	34	5	34	5

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	50	33	50	32

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17
Outplane Stress Summary	32	9	32	9

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	40	34	40
Inplane Stress Summary	51	28	51	28

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Vessel Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	51	29	51	28
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	39	46	39	46

Hoop Stress Summary	43	50	43	50
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	39	-39	39	-39
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	19	19	19	19

Axial Stress Summary	62	-15	62	-16
----------------------	----	-----	----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	4	4	-4	-4
----------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	63	66	63	67
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	39	46	39	46



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Hoop Stress Summary	43	50	43	50
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	39	-39	39	-39
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	19	19	19	19

Axial Stress Summary	61	-15	62	-16
----------------------	----	-----	----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-4	-4	4	4
----------------------	----	----	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	62	66	63	67
------------------------	----	----	----	----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{130.881(11.113 - 3.0)}$$

$$= 221.830 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	261.763 mm
Nozzle Outside Diameter	(d) =	200.000 mm
Vessel Thickness used	(T) =	8.113 mm
Nozzle Thickness used	(t) =	50.975 mm
T / t	=	0.159
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.340$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.047
Mr/P	=	0.076
M0/P	=	0.023
N0/P	=	0.043
MrD/Mc	=	0.250
NrDL/Mc	=	0.093
M0d/Mc	=	0.075
N0DL/Mc	=	0.073
MrD/Ml	=	0.061
NrDL/Ml	=	0.048
M0D/Ml	=	0.019



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NØDL/M1 = 0.044

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	89	95	89	95
Outplane Stress Summary	114	82	114	82

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	62	-62	62	-62
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	44	44	44	44
Inplane Stress Summary	113	-11	113	-11

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	2	2	-2	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	116	93	116	93

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	44	44	44	44
Outplane Stress Summary	69	31	69	31



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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	62	-62	62	-62
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	89	95	89	95
Inplane Stress Summary	158	39	159	39

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-2	-2	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	158	40	159	39

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	34	40	34	40	34	40	34	40
Circ. Pl (SUS)	3	3	3	3	2	2	2	2
Circ. Q (SUS)	11	-11	11	-11	14	-14	14	-14
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	2	2	2	2	3	3	3	3
Long. Q (SUS)	14	-14	14	-14	11	-11	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	34.0	40.0	34.0	40.0	34.0	40.0	34.0	40.0
Pm+Pl (SUS)	37.1	43.0	37.1	43.0	36.1	42.0	36.1	42.0
Pm+Pl+Q (Total)	48.1	32.0	48.1	32.0	50.1	28.1	50.1	28.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	40.00	98.98	Passed
Pm+Pl (SUS)	43.04	148.47	Passed
Pm+Pl+Q (TOTAL)	50.05	296.95	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	89	95	89	95	89	95	89	95
Circ. Pl (SUS)	5	5	5	5	6	6	6	6
Circ. Q (SUS)	19	-19	19	-19	62	-62	62	-62
Long. Pm (SUS)	44	44	44	44	44	44	44	44
Long. Pl (SUS)	6	6	6	6	5	5	5	5
Long. Q (SUS)	62	-62	62	-62	19	-19	19	-19
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	89.0	95.0	89.0	95.0	89.0	95.0	89.0	95.0
Pm+Pl (SUS)	94.1	100.1	94.1	100.1	95.1	101.1	95.1	101.1
Pm+Pl+Q (Total)	114.6	93.1	114.6	93.1	157.0	39.4	157.0	39.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	95.00	98.98	Passed
Pm+Pl (SUS)	101.08	148.47	Passed
Pm+Pl+Q (TOTAL)	157.04	296.95	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	39	46	39	46	39	46	39	46
Circ. Pl (SUS)	3	3	3	3	3	3	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	3	3	3	3	3	3	3	3
Long. Q (SUS)	39	-39	39	-39	39	-39	39	-39
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	39.0	46.0	39.0	46.0	39.0	46.0	39.0	46.0



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Nozzle Calcs.: N3-Shell inlet

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Pm+Pl (SUS)	42.8	49.6	42.8	49.6	42.8	49.6	42.8	49.6
Pm+Pl+Q (Total)	61.8	66.5	61.8	66.5	61.8	66.5	61.8	66.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	98.98	Passed
Pm+Pl (SUS)	49.58	148.47	Passed
Pm+Pl+Q (TOTAL)	66.48	296.95	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N4-Shell outlet

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Input, Nozzle Desc: N4-Shell outlet

From: 40

Pressure for Reinforcement Calculations	P	5904.901	KPa.
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	430	°C
Parent Material		SA-335 P11	
Parent Allowable Stress at Temperature	Sv	98.98	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	247.65	mm
Design Length of Section	L	2550.0002	mm
Shell Finished (Minimum) Thickness	t	11.1125	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2527.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	98.98	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.8231	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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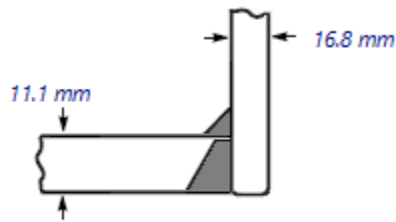
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Nozzle Calcs.: N4-Shell outlet

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Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5905 \cdot 127 / (99 \cdot 1 - 0.6 \cdot 5905)$
 $= 7.8473 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5905 \cdot 28.4 / (99 \cdot 1 - 0.6 \cdot 5905)$
 $= 1.7572 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4257 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.2812 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4-Shell outlet.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.7572 mm
Wall Thickness per UG16(b),	tr16b	= 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 10.8473 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 10.8473 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 5.5000 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(10.8, 5.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.76, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK



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Nozzle Calcs.: N4-Shell outlet

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Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	3.4,	Allowable	:	99.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	267.8 N./mm ²	Passed
Occasional	:	4.9,	Allowable	:	131.6 N./mm ²	Passed
Shear	:	3.8,	Allowable	:	69.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 11.1, tr = 6.54, c = 3 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.81$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $5904.90 / 15310.00 = 0.386$

Weld Size Calculations, Description: N4-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 9.7000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 7.3998 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 215.3623 mm

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.0000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	247.650 mm
Shell Thickness	11.1125 mm
Shell Stress Concentration Factor	1.000
Vessel Material	SA-335 P11
Vessel Cold S.I. Allowable	Smc 118.00 N./mm ²
Vessel Hot S.I. Allowable	Smh 98.98 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID
Nozzle Corrosion Allowance	3.0000 mm
Nozzle Diameter	50.800 mm
Nozzle Thickness	16.7640 mm
Nozzle Stress Concentration Factor	1.000



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Nozzle Calcs.: N4-Shell outlet

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Nozzle Material	SA-335 P11		
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	98.98	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	5904.90	KPa.
Radial Load	P	-458.90	Kgf
Circumferential Shear	Vc	331.43	Kgf
Longitudinal Shear	Vl	331.43	Kgf
Circumferential Moment	Mc	0.05	Kg-m.
Longitudinal Moment	Ml	0.05	Kg-m.
Torsional Moment	Mt	0.06	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.030) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{130.881(11.113 - 3.0)}$$

$$= 191.858 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	261.763 mm
Nozzle Outside Diameter	(d) =	84.328 mm
Vessel Thickness used	(T) =	8.113 mm
Nozzle Thickness used	(t) =	13.764 mm
T / t	=	0.589
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.830$$

Nr/P	=	0.075
Mr/P	=	0.146
M0/P	=	0.043
N0/P	=	0.102
MrD/Mc	=	0.281
NrDL/Mc	=	0.107
M0d/Mc	=	0.090
N0DL/Mc	=	0.064
MrD/Ml	=	0.121
NrDL/Ml	=	0.071
M0D/Ml	=	0.040
N0DL/Ml	=	0.128

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0



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Nozzle Calcs.: N4-Shell outlet

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Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	89	95	89	95

Outplane Stress Summary	114	84	114	84
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	44	44	44	44

Inplane Stress Summary	109	-9	109	-9
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	3	3	-3	-3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	115	94	115	94
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	44	44	44	44

Outplane Stress Summary	69	34	69	33
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0



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Nozzle Calcs.: N4-Shell outlet

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Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	89	95	89	95

Inplane Stress Summary	154	40	154	40
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-3	-3	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	154	42	154	41
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	18	11	18

Hoop Stress Summary	18	25	18	25
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	15	-15	15	-15
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	22	-8	22	-8
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	1	1	-1	-1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: N4-Shell outlet

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(Stress Intensities	Outside	Inside	Outside	Inside
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Two * Max Shear Stress	23	34	23	34
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	18	11	18

Hoop Stress Summary	18	25	18	25
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	15	-15	15	-15
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	22	-8	22	-8
----------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-1	-1	1	1
----------------------	----	----	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	23	34	23	34
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	89	95	89	95	89	95	89	95
Circ. Pl (SUS)	7	7	7	7	5	5	5	5
Circ. Q (SUS)	17	-17	17	-17	59	-59	59	-59
Long. Pm (SUS)	44	44	44	44	44	44	44	44
Long. Pl (SUS)	5	5	5	5	7	7	7	7
Long. Q (SUS)	59	-59	59	-59	17	-17	17	-17



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Nozzle Calcs.: N4-Shell outlet

Nozl: 11

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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	89.0	95.0	89.0	95.0	89.0	95.0	89.0	95.0
Pm+P1 (SUS)	96.2	102.2	96.2	102.2	94.2	100.2	94.2	100.2
Pm+P1+Q (Total)	114.4	95.2	114.4	95.2	153.1	42.1	153.1	42.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	95.00	98.98	Passed
Pm+P1 (SUS)	102.17	148.47	Passed
Pm+P1+Q (TOTAL)	153.11	296.95	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	18	11	18	11	18	11	18
Circ. P1 (SUS)	7	7	7	7	7	7	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. P1 (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	15	-15	15	-15	15	-15	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	11.0	18.0	11.0	18.0	11.0	18.0	11.0	18.0
Pm+P1 (SUS)	18.1	25.1	18.1	25.1	18.1	25.1	18.1	25.1
Pm+P1+Q (Total)	21.3	34.1	21.3	34.1	21.3	34.1	21.3	34.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	18.00	98.98	Passed
Pm+P1 (SUS)	25.05	148.47	Passed
Pm+P1+Q (TOTAL)	34.06	296.95	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2-Tube Outlet

Nozl: 12

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Input, Nozzle Desc: N2-Tube Outlet

From: 50

Pressure for Reinforcement Calculations	P	2402.754	KPa.
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	270	°C
Parent Material		SA-335 P11	
Parent Allowable Stress at Temperature	Sv	110.80	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	247.65	mm
Design Length of Section	L	300.0311	mm
Shell Finished (Minimum) Thickness	t	11.1125	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3227.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-335 P11	
Material UNS Number		K11597	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	110.80	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.1121	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.20	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	214.3000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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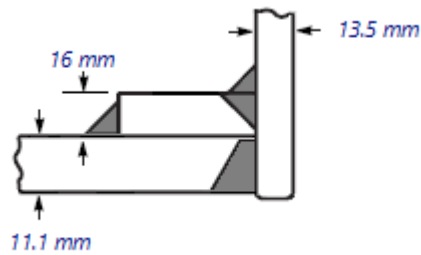
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Nozzle Calcs.: N2-Tube Outlet

Noz1: 12

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Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.438 in.
Actual Thickness Used in Calculation	0.531 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2403 \cdot 127 / (111 \cdot 1 - 0.6 \cdot 2403)$
 $= 2.7867 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.4491 and the shell thk. is 11.1125 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2403 \cdot 46.7 / (111 \cdot 1 - 0.6 \cdot 2403)$
 $= 1.0253 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4236 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	186.6504 mm
Parallel to Vessel Wall, opening length	d	93.3252 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		20.2812 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 111 / 111)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 111 / 111)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137 / 111)$
 $= 1.000$



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Nozzle Calcs.: N2-Tube Outlet

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Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(1, 1)

= 1.000

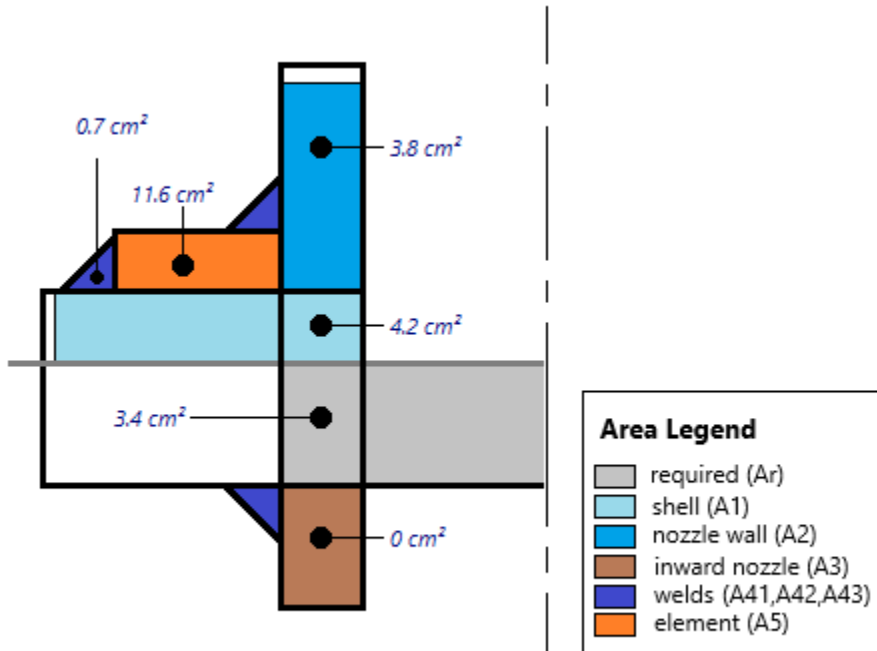


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3.400	0.436	NA
Area in Shell	A1	4.171	6.699	NA
Area in Nozzle Wall	A2	3.838	4.082	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.673	0.673	NA
Area in Element	A5	11.576	11.576	NA
TOTAL AREA AVAILABLE	Atot	20.258	23.030	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)

= (93.3 • 3.64 • 1 + 2 • 10.5 • 3.64 • 1 • (1-1))

= 3.400 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)

= 93.3 (1 • 8.11 - 1 • 3.64) - 2 • 10.5

(1 • 8.11 - 1 • 3.64) • (1 - 1)



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Nozzle Calcs.: N2-Tube Outlet

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$$= 4.171 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 20.3) \cdot (10.5 - 1.03) \cdot 1$$

$$= 3.838 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_{r \text{ Lost}}) \cdot f_{r3} + ((W_i - c_{an}/0.707)^2 - A_{r \text{ Lost}}) \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= (0.67)^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$$

$$= 0.673 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (187 - 114) \cdot 16 \cdot 1$$

$$= 11.576 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.0253 \text{ mm}$

Wall Thickness per UG16(b), $t_{r16b} = 5.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $t_{rb1} = 5.7867 \text{ mm}$

Wall Thickness $t_{b1} = \max(t_{rb1}, t_{r16b}) = 5.7867 \text{ mm}$

Wall Thickness $t_{b2} = \max(t_{rb2}, t_{r16b}) = 5.5000 \text{ mm}$

Wall Thickness per table UG-45 $t_{b3} = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[8.26, \max(5.79, 5.5)]$$

$$= 5.7867 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.03, 5.79)$$

$$= 5.7867 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 13.487 = 11.801 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	36.9,	Allowable	:	110.8 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	249.1 N./mm ²	Passed
Occasional	:	4.8,	Allowable	:	147.4 N./mm ²	Passed
Shear	:	22.0,	Allowable	:	77.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.8$, $t_r = 0.96$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.11$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A -3°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 11.8$, $t_r = 0.96$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.11$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B -24°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C



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Nozzle Calcs.: N2-Tube Outlet

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Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 11.8$, $t_r = 0.96$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-24 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 11.1$, $t_r = 2.61$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.32$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-42 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 11.1$, $t_r = 2.61$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.32$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $2402.75 / 10200.00 = 0.236$

Weld Size Calculations, Description: N2-Tube Outlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	10.4874 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	9.7000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$4.8500 = 0.5 \cdot T_{minPad}$	$8.4840 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3.4 - 4.17 + 2 \cdot 10.5 \cdot 1 \cdot (1 \cdot 8.11 - 3.64)) \cdot 111)$
 $= 188.43 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_r2) \cdot S_v$
 $= (3.84 + 11.6 + 0.67 - 0 \cdot 1) \cdot 111$
 $= 18175.48 \text{ Kgf}$

Weld Load [W2]:



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Nozzle Calcs.: N2-Tube Outlet

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$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (3.84 + 0 + 1 + (1.7)) \cdot 111$$

$$= 7388.61 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (3.84 + 0 + 0.67 + 11.6 + (1.7)) \cdot 111$$

$$= 20097.96 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 111$$

$$= 9940. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 214 \cdot 12 \cdot 0.49 \cdot 111$$

$$= 22363. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 51.9) \cdot (13.5 - 3) \cdot 0.7 \cdot 111$$

$$= 13525. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 114 \cdot 16 \cdot 0.74 \cdot 137$$

$$= 29740. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 114 \cdot (8.11 - 0) \cdot 0.74 \cdot 111$$

$$= 12177. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (22363 + 13525) = 35888 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9940 + 29740 + 12177 + 0) = 51856 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (22363 + 12177 + 0) = 34540 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 35888	Kgf, must exceed W = 188 Kgf	or W1 = 18175 Kgf
Path 2-2 = 51856	Kgf, must exceed W = 188 Kgf	or W2 = 7388 Kgf
Path 3-3 = 34539	Kgf, must exceed W = 188 Kgf	or W3 = 20097 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6821 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 13.9773 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 175.0898 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:



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Nozzle Calcs.: N2-Tube Outlet

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Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	247.650	mm
Shell Thickness	11.1125	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-335 P11	
Vessel Cold S.I. Allowable	Smc 118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 110.80	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	87.325	mm
Nozzle Thickness	13.4874	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-335 P11	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 110.80	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	214.300	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	2402.75	KPa.
Radial Load	P	-917.81	Kgf
Circumferential Shear	Vc	662.86	Kgf
Longitudinal Shear	Vl	662.86	Kgf
Circumferential Moment	Mc	203.88	Kg-m.
Longitudinal Moment	Ml	203.88	Kg-m.
Torsional Moment	Mt	234.46	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (8.475) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{138.881(11.113 - 3.0)}$$

$$= 225.068 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	277.763	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	24.112	mm
Nozzle Thickness used	(t) =	10.487	mm
T / t	=	2.299	
d / t	=	10.899	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.397$$



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Nozzle Calcs.: N2-Tube Outlet

Nozl: 12

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Nr/P = 0.138
 Mr/P = 0.119
 MØ/P = 0.084
 NØ/P = 0.166
 MrD/Mc = 0.200
 NrDL/Mc = 0.184
 MØd/Mc = 0.163
 NØDL/Mc = 0.056
 MrD/Ml = 0.155
 NrDL/Ml = 0.145
 MØD/Ml = 0.131
 NØDL/Ml = 0.170

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	11	13	11	13
Outplane Stress Summary	-6	27	50	-9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	5	5	5	5
Inplane Stress Summary	-13	20	51	-26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	17	30	54	27

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	D1 Right
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circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-29	29	29	-29
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	5	5	5	5

Outplane Stress Summary	-15	28	47	-27
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-36	36	36	-36
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	11	13	11	13

Inplane Stress Summary	-16	35	66	-25
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	19	36	68	32

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	14	11	14

Hoop Stress Summary	8	11	19	22
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	2	2	2	2
Axial Bending (P)	24	-24	24	-24
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-24	-24	24	24
Axial Bending (ML)	-79	79	79	-79
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-70	38	136	-70

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	14	14	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	84	44	137	93

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	14	11	14
Hoop Stress Summary	12	15	15	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	24	-24	24	-24
Axial Membrane (Mc)	-24	-24	24	24
Axial Bending (Mc)	-102	102	102	-102
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-94	62	160	-94

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	10	10	10	10
Shear Stress Summary	7	7	14	14



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	107	63	161	116

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{130.881(11.113 - 3.0)}$$

$$= 221.830 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	261.763 mm
Nozzle Outside Diameter	(d) =	214.300 mm
Vessel Thickness used	(T) =	8.113 mm
Nozzle Thickness used	(t) =	60.487 mm
T / t	=	0.134
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.650$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.046
Mr/P	=	0.072
M0/P	=	0.022
N0/P	=	0.037
MrD/Mc	=	0.246
NrDL/Mc	=	0.094
M0d/Mc	=	0.074
N0DL/Mc	=	0.066
MrD/Ml	=	0.056
NrDL/Ml	=	0.047
M0D/Ml	=	0.017
N0DL/Ml	=	0.036

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	36	38	36	38
Outplane Stress Summary	39	35	79	16

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



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Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-47	47	47	-47
Inplane Pressure Stress	18	18	18	18

Inplane Stress Summary	28	6	137	-75
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	6	6	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	42	37	137	92
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-62	62	62	-62
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18	18	18	18

Outplane Stress Summary	-30	58	112	-47
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-209	209	209	-209
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	36	38	36	38

Inplane Stress Summary	-121	182	324	-209
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	121	182	324	209

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	13	11	13	11	13	11	13
Circ. Pl (SUS)	-3	-3	7	7	-3	-3	7	7
Circ. Q (SUS)	-16	16	30	-30	-25	25	47	-47
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-2	-2	6	6	1	1	3	3
Long. Q (SUS)	-17	17	39	-39	-22	22	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	11.0	13.0	11.0	13.0	11.0	13.0	11.0	13.0
Pm+Pl (SUS)	8.2	10.1	18.1	20.1	8.4	10.2	18.1	20.1
Pm+Pl+Q (Total)	16.8	28.8	52.2	28.5	19.5	36.1	66.1	32.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	110.80	Passed
Pm+Pl (SUS)	20.11	166.20	Passed
Pm+Pl+Q (TOTAL)	66.13	332.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	36	38	36	38	36	38	36	38
Circ. Pl (SUS)	0	0	10	10	-7	-7	19	19
Circ. Q (SUS)	3	-3	31	-31	-151	151	267	-267
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	0	0	12	12	-4	-4	14	14
Long. Q (SUS)	11	-11	105	-105	-45	45	79	-79
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	36.0	38.0	36.0	38.0	36.0	38.0	36.0	38.0



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Pm+Pl (SUS)	36.2	38.2	46.2	48.2	29.3	31.2	55.2	57.2
Pm+Pl+Q (Total)	41.1	35.9	135.0	92.0	122.0	182.0	322.1	210.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.00	110.80	Passed
Pm+Pl (SUS)	57.16	166.20	Passed
Pm+Pl+Q (TOTAL)	322.12	332.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	14	11	14	11	14	11	14
Circ. Pl (SUS)	-3	-3	7	7	1	1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-22	-22	26	26	-22	-22	26	26
Long. Q (SUS)	-55	55	103	-103	-78	78	126	-126
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	11.0	14.0	11.0	14.0	11.0	14.0	11.0	14.0
Pm+Pl (SUS)	25.7	28.6	31.7	31.8	29.6	32.6	31.5	31.6
Pm+Pl+Q (Total)	84.1	43.2	134.4	94.0	107.9	62.0	158.2	115.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	110.80	Passed
Pm+Pl (SUS)	32.56	166.20	Passed
Pm+Pl+Q (TOTAL)	158.17	332.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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ASME TS Calc:

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Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	5900.00 KPa.
Shell Maximum Operating Pressure	Psox,max	5900.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	12.7000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	247.650 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	430.00 °C
Shell Material		SA-335 P11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11597
Shell Allowable Stress at Temperature	Ss	98.98 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: Bottom ch shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	2400.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	2400.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	12.7000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	247.650 mm
Channel Design Temperature	TEMPC	270.00 °C
Channel Flange Matl		SA-335 P11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11597
Channel Allowable Stress at Temperature	Sc	110.80 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	60
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	2650.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2550.00 mm
Design Temperature of the Tubes		270.00 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		110.80 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	166.60 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	101.600 mm



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ASME TS Calc:

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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	268.2 cm ²
Perimeter of Tube Layout	Cp	609.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	270.00 °C
Tubesheet Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K11789
Tubesheet Allowable Stress at Temperature	S	148.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00 N./mm ²
Thickness of Tubesheet	h	50.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	269.278 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	370.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Ps _{d,min}	-103.4190 KPa.
Channel Side Min. Design Pressure	P _t _{d,min}	-103.4190 KPa.
Mean Shell Metal Temp. along Shell len.	T _{sm}	0.00 °C
Mean Tube Metal Temp. along Tube length	T _{tm}	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



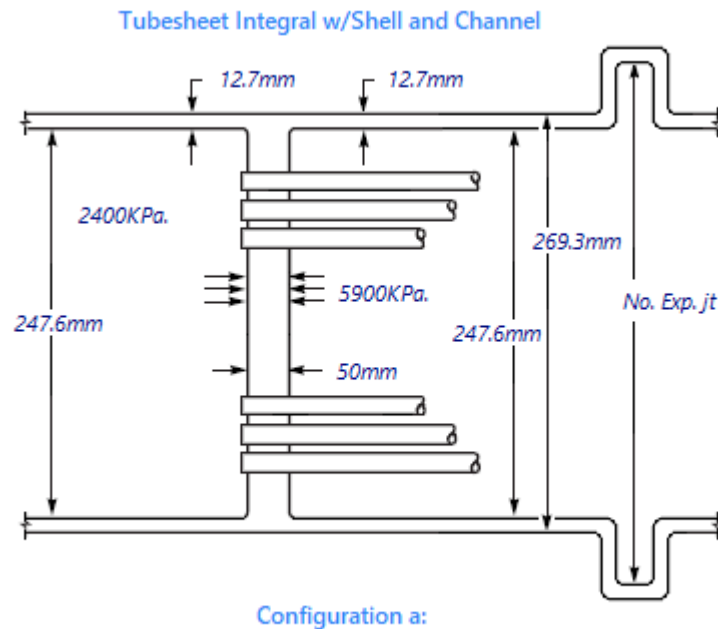
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ASME TS Calc:

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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	430.0 °C	0.17617E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20547E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	270.0 °C	0.18829E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	270.0 °C	0.18829E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20547E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	270.0 °C	0.18829E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	270.0 °C	0.18829E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Note:



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The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (2503 \cdot (19/2 - 0)) / (111 \cdot 1 + 0.4 \cdot 2503)$$

$$= 0.2133 + 0.0000 = 0.2133 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 270.00 °C
Elastic Modulus for Material 184606480.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	2550.00	9.03	50.0000	0.0134948	104.87
EMAWP = (2.167 / (D/T) - 0.0833) * B = 16435 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.0180	19.05	2550.00	18.71	50.0000	0.0031413	84.27
EMAWP = (4 * B) / (3 * (D/T)) = (4 * 84.3) / (3 * 18.7) = 6004 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.0180 mm
Allowable Internal Pressure at Corroded thickness	26929.28 KPa.
Required Internal Design Pressure	2503.42 KPa.
Allowable External Pressure at Corroded thickness	16434.91 KPa.
Required External Design Pressure	6003.42 KPa.
Required Thickness due to Shell Side pressure	1.0180 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/50$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(19 - 2 \cdot 2.11 (188292960/188292960) \cdot (111/148) \cdot (0); 19 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 102 + 19$$

$$= 222.250 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (25.4 - 19) / 25.4$$

$$= 0.250$$



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Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= D_o / 2 \\ &= 222 / 2 \\ &= 111.1250 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) } \\ &= 25.4 / \sqrt{1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 222 \cdot 25.4) / (\pi \cdot 222^2) } \\ &= 25.4000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (25.4 - 19) / 25.4 \\ &= 0.2500 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= a_s / a_o \\ &= 124 / 111 \\ &= 1.1143 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= a_c / a_o \\ &= 124 / 111 \\ &= 1.1143 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2 \\ &= 1 - 60 ((19 - 2 \cdot 2.11) / (2 \cdot 111))^2 \\ &= 0.7329 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot a_o))^2 \\ &= 1 - 60 (19 / (2 \cdot 111))^2 \\ &= 0.5592 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned} &= \max((h_g - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((0 - 0); 0) \\ &= 0.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$\begin{aligned} &= \pi \cdot ts(D_s + ts) E_s / L \\ &= \pi \cdot 12.7 (248 + 12.7) 176174368 / 2550 \\ &= 73183.0000 \text{ Kg/mm} \end{aligned}$$

Axial Tube Stiffness [Kt]:

$$\begin{aligned} &= \pi \cdot tt(D_t - tt) E_t / L \\ &= \pi \cdot 2.11 (19 - 2.11) 188292960 / 2550 \\ &= 845.5438 \text{ Kg/mm} \end{aligned}$$

Stiffness Factor [Ks,t]:

$$\begin{aligned} &= K_s / (Nt \cdot K_t) \\ &= 73183 / (60 \cdot 846) \\ &= 1.44252 \end{aligned}$$

Rigidity Ratio [J]:

$$\begin{aligned} &= 1 / (1 + K_s / K_j) \\ &= 1 / (1 + 73183 / 0) \\ &= 1.0000 (= 1 \text{ if No Exp. Jt.}) \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts) ts) \\ &= ((12(1 - 0.3^2))^{1/4}) / ((248 + 12.7) 12.7) \end{aligned}$$



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$$= 0.0316 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.032 \cdot 176174368 \cdot 12.7^3 / (6(1 - 0.3^2))$$

$$= 2089470336.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(248 + 12.7) \cdot 2089470336 / 50^3 \cdot (1 + 50 \cdot 0.032 + (50 \cdot 0.032)^2 / 2)$$

$$= 100006848.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 248^2 / (4 \cdot 176174368 \cdot 12.7) \cdot (1 - 248 \cdot 0.3 / (2(248 + 12.7)))$$

$$= 0.58747\text{E-}02 \text{ mm/N./mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(248 + 12.7)12.7}$$

$$= 0.0316 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.032 \cdot 188292960 \cdot 12.7^3 / (6(1 - 0.3^2))$$

$$= 2233199872.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(248 + 12.7 / \cos(0)) \cdot 2.23\text{E}+09 / 50^3 \cdot (1 + 50 \cdot 0.032 \cdot (1 + 1) / 2 + (50 \cdot 0.032)^2 / 2 \cdot 1)$$

$$= 106886072.0 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 248^2 / (4 \cdot 188292960 \cdot 12.7) \cdot (1 - 248 \cdot 0.3 / (2(248 + 12.7)))$$

$$= 0.54966\text{E-}02 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 1.968504 ; \mu^* = 0.250000 ; E^* = 40985988 \text{ KPa}$$

$$E^*/E = 0.217671 ; \nu^* = 0.395112$$

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t)_{ao^2}) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.4^2)60 \cdot 188292960 \cdot 2.11(19 - 2.11)^{111^2}) / (40985988 \cdot 2550 \cdot 50^3))^{1/4}$$

$$= 1.6675$$

Values from Table 4.18.3

$$Z_d = 0.309596 ; Z_v = 0.166405 ; Z_m = 0.678824$$

$$Z_a = 0.638921\text{E}+00 ; Z_w = 0.166405$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 269/222$$



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ASME TS Calc:

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$$= 1.2116$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.4) / 40985988 \cdot (100006848 + 106886072 + 188292960 \cdot \ln(1.21)) \\ &= 3.5868 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu) F \\ &= (1 + 0.4) 3.59 \\ &= 5.0040 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.11 - 1 - 5 \cdot 0.17) / (1 + 5 \cdot 0.68) \\ &= -0.163391 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4 \\ &= (0.31 + -0.16 \cdot 0.17) / 2 \cdot 1.67^4 \\ &= 1.0916 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4 \\ &= (0.17 + -0.16 \cdot 0.68) / 2 \cdot 1.67^4 \\ &= 0.2145 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m) \\ &= (0.17 + (1.11 - 1) 0.68) 1.67^4 / (1 + 5 \cdot 0.68) \\ &= 0.4290 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ &= 1.11 \cdot 2089470336 \cdot 0.032 \cdot 0.005874711 (1 + 50 \cdot 0.032) \\ &= 1115.9744 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω^*s]:

$$\begin{aligned} &= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s \\ &= 111^2 (1.11^2 - 1) \cdot (1.11 - 1) / 4 - 1116 \\ &= -1030.7212 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)] \\ &= 1.11 \cdot 2.23E+09 [0.005496613 (1 + 0.032 \cdot 1 \cdot 50) - 0.000 (1/50 + 1 \cdot 0.032/2)] \\ &= 1115.9744 \text{ mm}^2 \end{aligned}$$

Parameter [ω^*c]:

$$\begin{aligned} &= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c \\ &= 111^2 [(1.11^2 + 1) \cdot (1.11 - 1) / 4 - (1.11 - 1) / 2] - 1116 \\ &= -1030.7212 \text{ mm}^2 \end{aligned}$$

Pressure [P^*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$



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Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot K_{st})] - [(1 - J)/(2J \cdot K_{st})] [(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 5900 \{ 0.56 + 2(1 - 0.56)0.3 + [2/1.44 \cdot (248/222)^2]0.3 - [(1.11^2 - 1)/(1 \cdot 1.44)] - [(1 - 1)/(2 \cdot 1 \cdot 1.44)] [(0^2 - (248)^2)/222^2] \}$$

$$= 6918.3882 \text{ KPa.}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_{st})] P_t$$

$$= [0.73 + 2(1 - 0.73)0.3 + 1/(1 \cdot 1.44)] 2400$$

$$= 3807.2915 \text{ KPa.}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 60 \cdot 846 \cdot 0 / (3.14 \cdot 111^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 0.43 \cdot 0 / (2 \cdot \pi \cdot 111^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_t)$$

$$= -(0.43/111^2) \cdot (-1031 \cdot 5900 - -1031 \cdot 2400)$$

$$= 125.3207 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_c)$$

$$= (0.43/111^2) \cdot (1116 \cdot 0 - 1116 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot K_{st} / (1 + J \cdot K_{st} (Q_{z1} + (p_s - 1)Q_{z2})) \cdot (P_s - P_t + P_y + P_w + P_w + P_{rim})$$

$$= 0.1000E+01 \cdot 1.44 / (1 + 1 \cdot 1.44 (1.09 + (1.11 - 1)0.21)) \cdot (6918 - 3807 + 0 + 0 + -0 + 125)$$

$$= 1788.7346 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega_s \cdot P_s - \omega_c \cdot P_t) - (\omega_s \cdot P_s - \omega_c \cdot P_c)) + W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-1031 \cdot 5900 - -1031 \cdot 2400) - (1116 \cdot 0 - 1116 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 5 \cdot 0.68)$$

$$= -83.6693 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2)$$

$$= -0.16 + 2 \cdot -83.7 / (1789 \cdot 111^2)$$

$$= -0.23768$$

Fm Value from Table 4.18.3 = 0.118840

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot a_o / (H - h_g))^2 \cdot P_e$$



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ASME TS Calc:

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$$= (1.5 \cdot 0.12 / 0.25) \cdot (2 \cdot 111 / (50 - 0))^2 \cdot 1789$$

$$= 25.2017 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 148 = 222 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.063 / 0.25) \cdot (2 \cdot 111 / (9.52 - 0))^2 \cdot 1071$$

$$= 221.9908 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 9.52 + 0 + 0$$

$$= 9.5243 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.25 \cdot 50 / 111) \cdot 118$$

$$= 26.6362 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 30.9755 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(9.52; 3.81)$$

$$= 9.5243 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.92, F_{tmax} = 1.04$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((5900 \cdot 0.56 - 2400 \cdot 0.73) - (1789) \cdot 0.92) / (0.73 - 0.56)$$

$$= -0.5844 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((5900 \cdot 0.56 - 2400 \cdot 0.73) - (1789) \cdot 1.04) / (0.73 - 0.56)$$

$$= -1.8816 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 1.88 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 110.7994 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.31 + -0.24 \cdot 0.17)1.67^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2)$$



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ASME TS Calc:

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Determine the factor [rt]:

$$= \left(\sqrt{\left(\frac{dt^2}{4} + \left(\frac{dt}{2} - 2 \cdot \frac{tt^2}{4} \right)^2} \right) \right) / 4$$

$$= \left(\sqrt{\left(\frac{19^2}{4} + \left(\frac{19}{2} - 2 \cdot \frac{2.11^2}{4} \right)^2} \right) \right) / 4$$

$$= 6.0355 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{\left(\frac{2 \cdot \pi^2 \cdot Et}{Syt} \right)}$$

$$= \sqrt{\left(\frac{2 \cdot \pi^2 \cdot 188292960}{167} \right)}$$

$$= 149.3679$$

Determine the factor [Ft]:

$$= k \cdot L/r$$

$$= 1 \cdot 370/6.04$$

$$= 61.3042$$

Buckling Allowable Stress [Stb]:

$$= (Sy/tFs) \cdot (1 - Ft/(2 \cdot Ct))$$

$$= (167/2) \cdot (1 - 61.3/(2 \cdot 149))$$

$$= 66.206 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 1.88 \cdot 1.12$$

$$= 21.55 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (19 - 2.11) 2.11$$

$$= 1.123 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.11 (19 - 2.11) 111$$

$$= 1268.687 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 111$$

$$= 1268.6866 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 111/111$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 111$$

$$= 842.9230 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{\left(\frac{0.75 \cdot do}{2} \right)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{\left(\frac{0.75 \cdot 19}{2} \right)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) 1 \cdot 1} - 0.75 \cdot 19 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm



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Tube-Tubesheet Jt allowable, 1269 is >= tube strength 1269 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 99 \cdot 1 = 99 \text{ N./mm}^2$$

Axial Membrane Stress in Shell $[\sigma_s, m]$:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s) \\ = 111^2 / ((248 + 12.7) 12.7) \cdot [1789 + (1.11^2 - 1)(5900 - 2400)] + 124^2 \cdot 2400 / ((248 + 12.7) 12.7) \\ = 20.9695 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 12.7 mm, the minimum Shell length adjacent to the tubesheet should be 100.947 mm

Shell Membrane Stress due to Joint Interaction $[\sigma_s, m]$:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s) \\ = 111^2 / ((248 + 12.7) 12.7) [1789 + (1.11^2 - 1)(5900 - 2400)] + 124^2 \cdot 2400 / ((248 + 12.7) 12.7) \\ = 20.9695 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction $[\sigma_s, b]$:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E^*) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \} \\ = 6 \cdot 2089470336 / 12.7^2 \{ 0.032 [0.005874711 \cdot 5900 + 124^2 \cdot 0 / (2.24E+09)] + 6(1 - 0.4^2) / (40985988) (111 / 50)^3 (1 + 50 \cdot 0.032 / 2) [1789 (0.17 + 0.68 \cdot -0.16) + 2 / 111^2 \cdot 0.68 \cdot -83.7] \} \\ = 86.8899 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s \\ \therefore |21| + |86.9| \leq 148 \text{ N./mm}^2 \\ \therefore 108, \text{ must be } \leq 148 \text{ N./mm}^2$$

Note:

For a given Channel thickness of 12.7 mm, the minimum Channel length adjacent to the tubesheet should be 100.947 mm

Channel Membrane Stress due to Joint Interaction $[\sigma_c, m]$:

$$= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta)) \\ = 124^2 \cdot 2400 / ((248 + 12.7) 12.7 \cdot \cos(0)) \\ = 11.130 \text{ N./mm}^2$$

Channel Bending Stress due to Joint Interaction $[\sigma_c, b]$:

$$= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \Theta) \cdot P_t - 6(1 - \nu^2) / (E^*) \cdot (a_o / h)^3 \cdot (C_m \Theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \} \\ = 6 \cdot 2.23E+09 / 12.7^2 \cdot \{ (0.005496613 \cdot 1 \cdot 0.032 - 0.000 \cdot 1) \cdot 2400 - 6(1 - 0.4^2) / 40985988 \cdot (111 / 50)^3 \cdot (1 + 50 \cdot 0.032 / 2 \cdot 1) \cdot [1789 (0.17 + 0.68 \cdot -0.16) + 2 / 111^2 \cdot 0.68 \cdot -83.7] \} \\ = 32.8237 \text{ N./mm}^2$$

Channel Stress Summation vs. Allowable

$$\therefore |\sigma_c, m| + |\sigma_c, b| \leq 1.5 \cdot S_c$$



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ASME TS Calc:

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:: | 11.1 | + | 32.8 | ≤ 166 N./mm²
 :: 44 must be ≤ 166 N./mm²

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	25.2 <=	222.0 N./mm ²	Pass
Tubesheet Shear Stress	6.3 <=	118.4 N./mm ²	Pass
Maximum Tube Stress	1.9 <=	110.8 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-1.9 <=	-66.2 N./mm ²	Pass
Maximum Force on any one Tube	21.5 <=	1268.7 Kgf	Pass
Axial Membrane Stress in Shell	21.0 <=	99.0 N./mm ²	Pass
Shell Stress (jt. inter.)	107.9 <=	148.5 N./mm ²	Pass
Channel Stress (jt. inter.)	44.0 <=	166.2 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	9.524	50.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 16 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Case Type	Pass/ Fail
D1uc	11.762	...	27	222	8 118 Ps+Pt-Th	D1 Ok
D2uc	21.043	...	53	222	14 118 Ps+Pt-Th	D2 Ok
D3uc	9.524	...	25	222	6 118 Ps+Pt-Th	D3 Ok
D4uc	3.810	...	...	222	...	118 Ps+Pt-Th D4 Ok
O1uc	4.293	...	26	500	8 118 Ps+Pt+Th	O1 Ok
O2uc	9.177	...	51	500	14 118 Ps+Pt+Th	O2 Ok
O3uc	3.810	...	25	500	6 118 Ps+Pt+Th	O3 Ok
O4uc	3.810	...	...	500	...	118 Ps+Pt+Th O4 Ok
D1c	17.234	...	36	222	9 118 Ps+Pt-Th-c	D1 Ok
D2c	24.821	...	59	222	15 118 Ps+Pt-Th-c	D2 Ok
D3c	14.349	...	26	222	6 118 Ps+Pt-Th-c	D3 Ok
D4c	9.810	...	1	222	...	118 Ps+Pt-Th-c D4 Ok
O1c	10.506	...	35	500	8 118 Ps+Pt+Th-c	O1 Ok
O2c	14.568	...	57	500	15 118 Ps+Pt+Th-c	O2 Ok
O3c	9.810	...	26	500	6 118 Ps+Pt+Th-c	O3 Ok
O4c	9.810	...	...	500	...	118 Ps+Pt+Th-c O4 Ok
Max:	24.8208	... mm	0.264	0.126	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance



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ASME TS Calc:

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[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	1	98	...	...		...	...	...	...	Ok
D2uc	20	98	...	...		...	...	...	...	Ok
D3uc	21	98	...	...		...	...	...	...	Ok
D4uc	...	98	...	-76		...	...	...	...	Ok
O1uc	1	296	...	...		...	...	...	...	Ok
O2uc	20	296	...	...		...	...	...	...	Ok
O3uc	21	296	...	...		...	...	...	...	Ok
O4uc	...	296	...	...		...	...	...	...	Ok
D1c	1	98	...	...		...	...	...	...	Ok
D2c	26	98	...	...		...	...	...	...	Ok
D3c	28	98	...	...		...	...	...	...	Ok
D4c	1	98	-1	-73		...	...	...	...	Ok
O1c	1	296	...	...		...	...	...	...	Ok
O2c	26	296	...	...		...	...	...	...	Ok
O3c	28	296	...	...		...	...	...	...	Ok
O4c	...	296	...	...		...	...	...	...	Ok
Max RATIO	0.280		0.007			...		...		

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Stress Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	3	110	...	...	36	1268	13	148	58	166	Ok
D2uc	5	110	-5	-66	58	1268	118	148	16	166	Ok
D3uc	2	110	-2	-66	22	1268	108	148	44	166	Ok
D4uc	...	110	...	-66	1	1268	2	148	2	166	Ok
O1uc	3	221	...	...	35	2537	12	296	57	333	Ok
O2uc	5	221	-5	-66	57	2537	118	296	13	333	Ok
O3uc	2	221	-2	-66	22	2537	108	296	44	333	Ok
O4uc	...	221	...	...	...	2537	...	296	...	333	Ok
D1c	3	110	...	-66	39	1268	26	148	86	166	Ok
D2c	4	110	-4	-66	49	1268	172	296	35	166	Ok
D3c	1	110	-1	-66	13	1268	150	296	55	166	Ok
D4c	...	110	...	-66	1	1268	2	148	3	166	Ok
O1c	3	221	...	-66	38	2537	24	296	86	333	Ok
O2c	4	221	-4	-66	48	2537	171	296	31	333	Ok
O3c	1	221	-1	-66	13	2537	150	296	55	333	Ok
O4c	...	221	...	...	...	2537	...	296	...	333	Ok
Max RATIO	0.046		0.077		0.046		0.796		0.520		

Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	24.821	50.000	Ok
Tubesheet thickness flanged extension :	0.000	14.000	Ok



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ASME TS Calc:

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Tube thickness :	1.018	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (12.700) adj. to tubesheet: 100.947 mm

Min Channel length of thk, (12.700) adj. to tubesheet: 100.947 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm

Due to Pressure Shell Compresses by : -0.095 mm

Due to Pressure + Thermal Shell Compresses by : -0.095 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	15041.83	0.988	22933.04	1.000
Tubesheet Shear Stress	33857.88	1.000	47810.27	1.000
Tube Tensile Stress	79397.89	1.000	132299.00	1.000
Tube Compressive Stress	No Calc	No Calc	79048.53	1.000
Tube-Tubesheet Joint load	79397.92	1.000	132292.55	1.000
Shell Stress (Axial, Junction)	13935.10	0.463	10248.45	1.000
Tubesheet-Channel Junction	9319.08	1.000	10248.45	0.453
Tube Pressure Stress	26929.27	1.000	16434.90	1.000
Minimum MAWP	9319.08		10248.45	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	20260.12	1.000	25275.68	1.000
Tubesheet Shear Stress	37612.14	1.000	50039.71	1.000
Tube Tensile Stress	93780.91	1.000	136642.16	1.000
Tube Compressive Stress	No Calc	No Calc	92740.73	1.000
Tube-Tubesheet Joint load	93780.91	1.000	136642.14	1.000
Shell Stress (Axial, Junction)	20260.12	0.252	19233.21	0.944
Tubesheet-Channel Junction	17238.04	1.000	19233.21	0.753
Tube Pressure Stress	28678.83	1.000	19233.21	1.000
Minimum MAPnc	17238.04		19233.21	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0125	0.0125	...	...	...	...
2	-0.0951	-0.0953	...	...	...	...
3	-0.0837	-0.0839	...	...	...	...
4	0.0012	0.0012	...	...	...	...
5	0.0109	0.0109	...	...	...	...
6	-0.0946	-0.0948	...	...	...	...
7	-0.0837	-0.0839	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:



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ASME TS Calc:

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Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned} &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\ &= \max(50/4, \min(50, 12.7)) \\ &= 12.700 \text{ mm} \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned} &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\ &= \min(5900/19233, 1) \\ &= 0.307 \end{aligned}$$

Thickness Ratio = 0.31, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$\begin{aligned} &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\ &= \max(50/4, \min(50, 12.7)) \\ &= 12.700 \text{ mm} \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned} &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\ &= \min(2400/17238, 1) \\ &= 0.139 \end{aligned}$$

Thickness Ratio = 0.14, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

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Problems/Failures Summary:

Step: 37

9:37am

Sep 9, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Channelside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: